

Communication and Hormone Balance

Addressing Adrenal Dysregulation and Chronic Stress



ROBERT LUBY, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

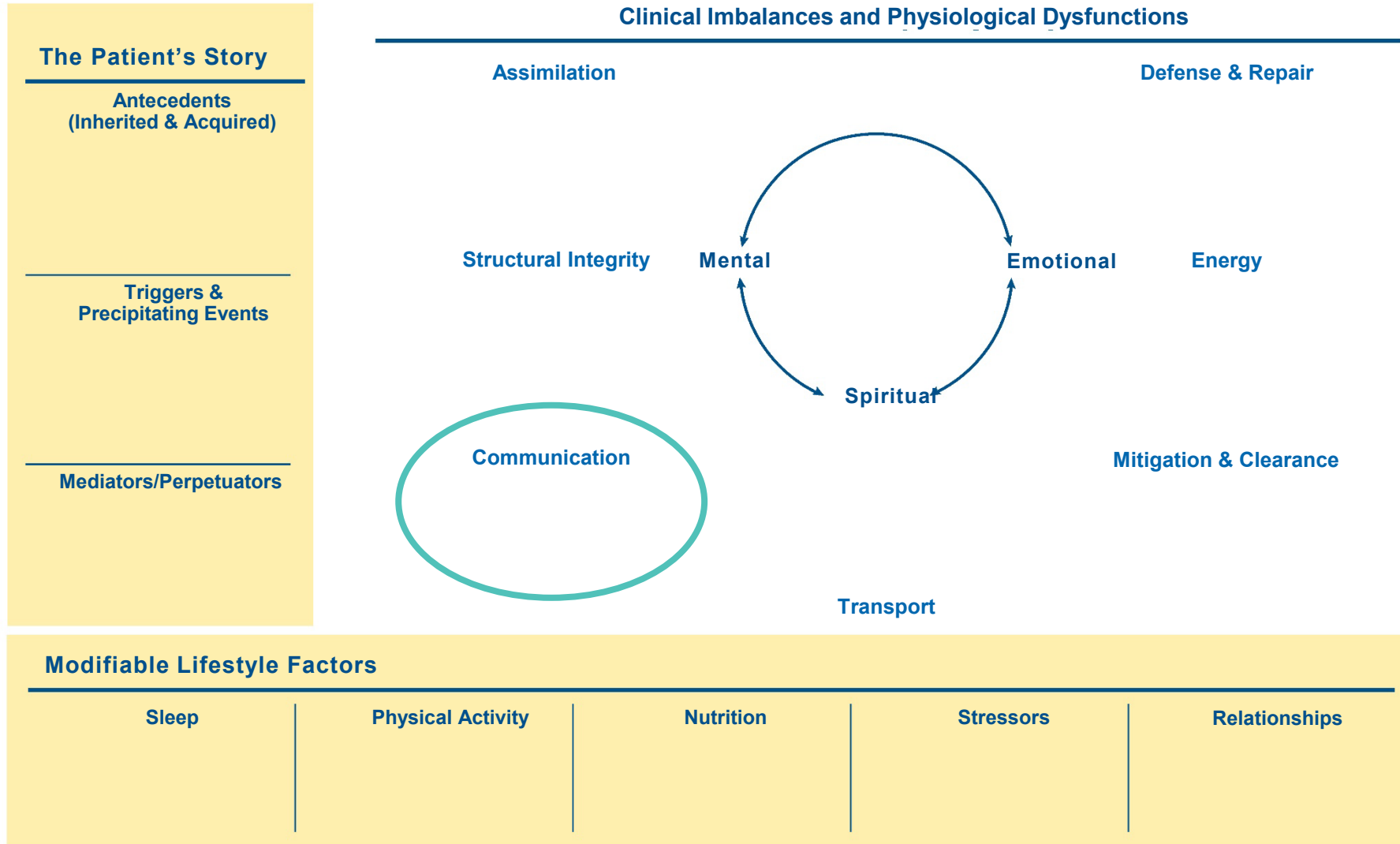
1. Explain key concepts and features that distinguish and explain the relationships among the following: acute stress, chronic stress, adrenal dysregulation, cortisol imbalance, allostasis, and allostatic load.
2. Cite clinical tools designed to identify and document root cause contributors (ATMs), clinical findings, and effects of chronic stress, adrenal dysregulation, and cortisol imbalance.
3. Explain the mechanisms by which dietary, lifestyle, mental, emotional, and spiritual factors and dysfunction in other bodily systems may contribute to adrenal dysregulation and cortisol imbalance.
4. Explain the mechanisms by which adrenal dysregulation and cortisol imbalance may cause dysregulation in other bodily systems

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document the patient's antecedents, triggers, mediators, and clinical findings of adrenal dysregulation, chronic stress, and cortisol imbalance on the IFM Timeline and Matrix.
2. Explain to a patient how chronic stress and other factors in their life (antecedents, triggers, and mediators) represent root causes of adrenal dysregulation and cortisol imbalance that underpin their clinical findings and current concerns.
3. Co-prioritize and initiate dietary, lifestyle, and gut health interventions as foundational therapies that address the patient's root cause contributors (ATMs) of stress, adrenal dysregulation, and cortisol imbalance and initiate foundational supplements as clinically indicated.

Functional Medicine Matrix



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

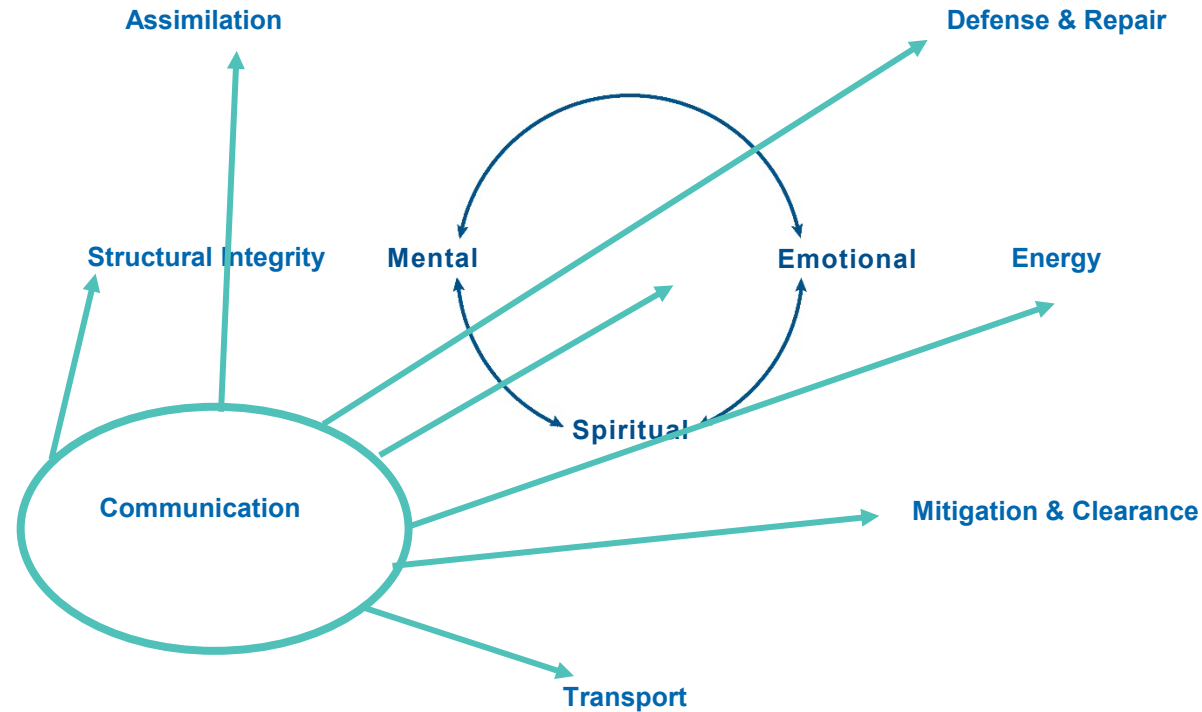
Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions



Systemic Effects
of Adrenal
Dysregulation
and the Chronic
Stress Response

Clinical Focus

- Overview of the acute stress vs. the chronic stress response.





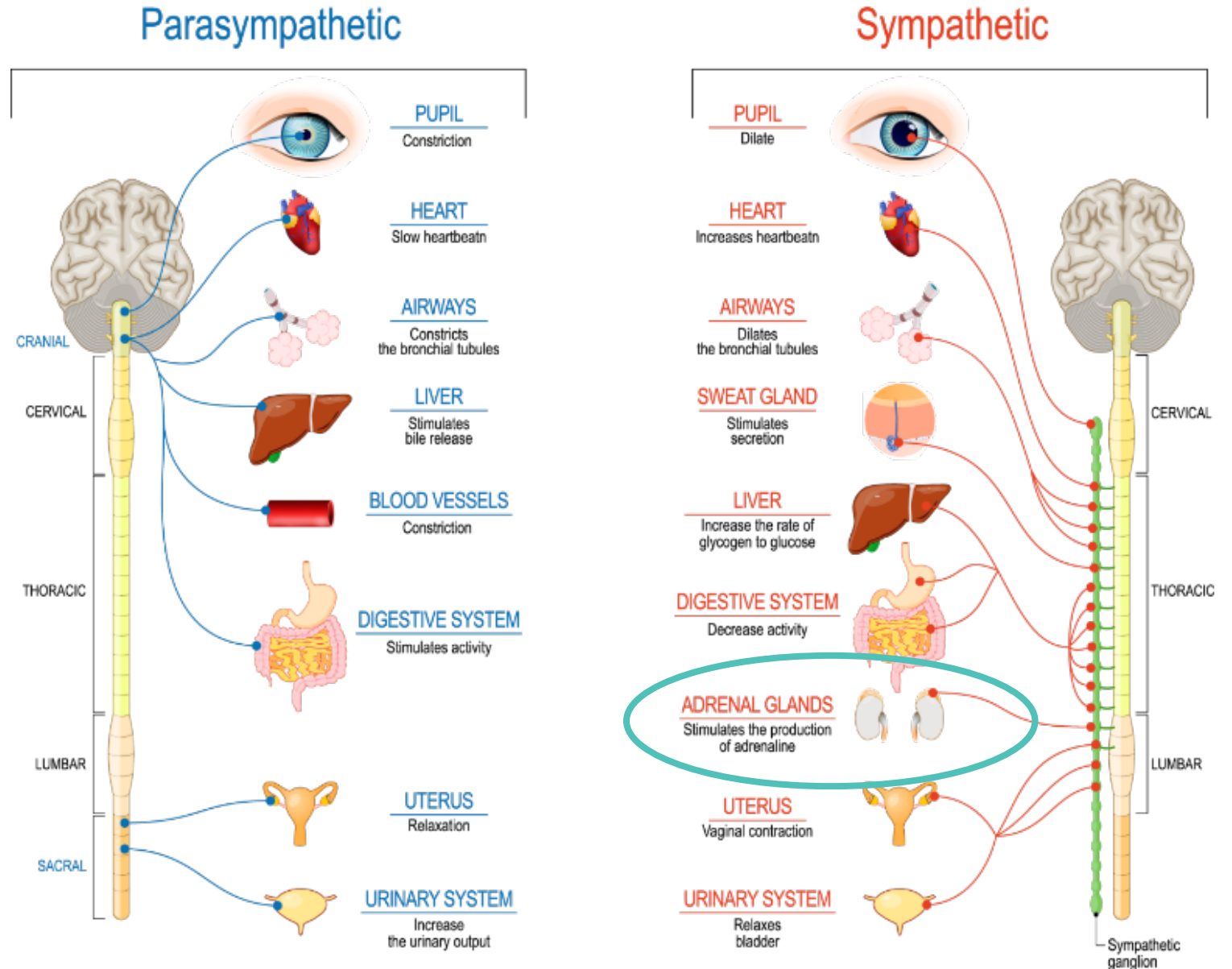
May Festival Chorus and Cincinnati Symphony Orchestra by KMoore CSymphony is licensed under CC BY 4.0.

Stress and the Cortisol Response



Nervous system

Overview of the PNS and ENS



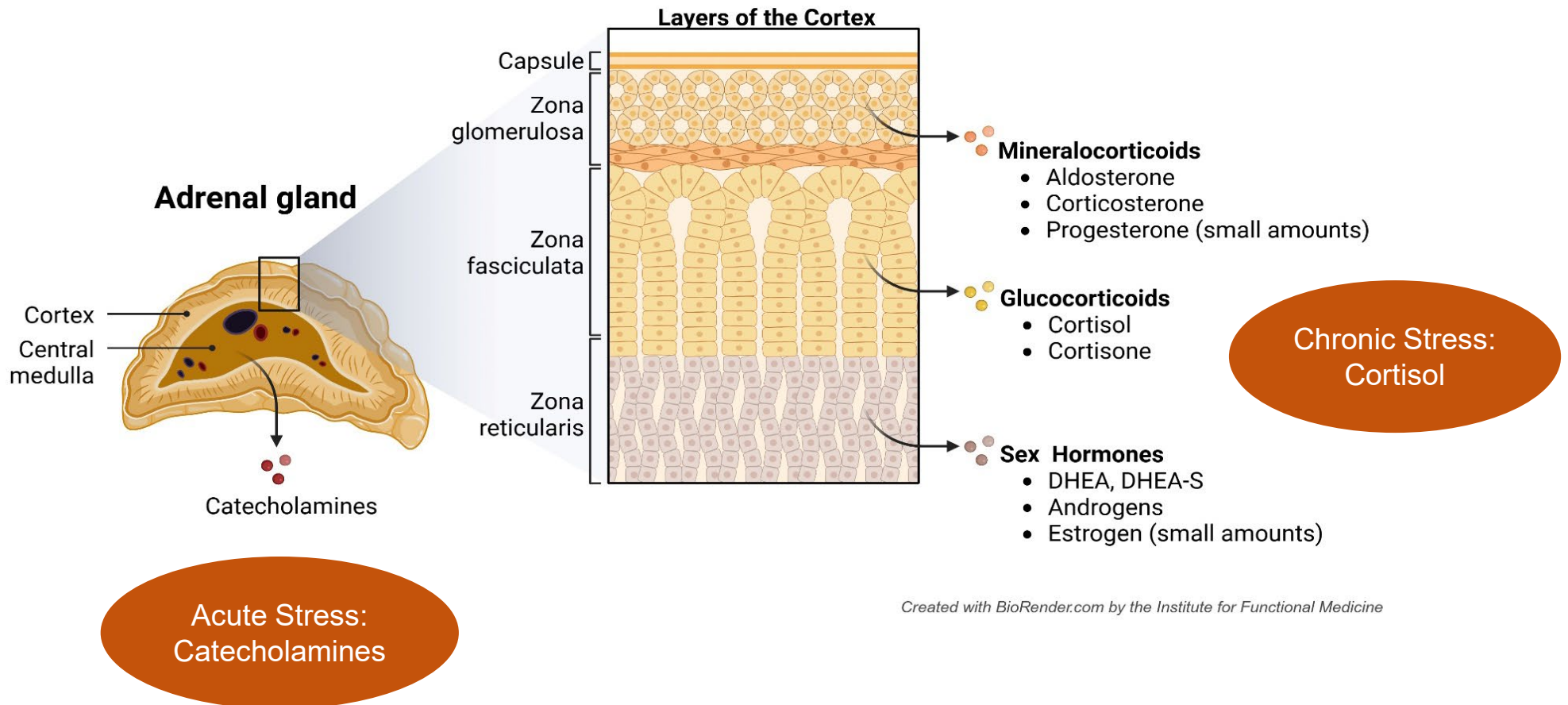
Stress and the Cortisol Response

Primarily met by two systems:
--Hypothalamic-pituitary-adrenal (**HPA**) axis
--Sympathetic nervous system (**SNS**)

Stressor-induced activation of the HPA axis and the SNS results in a series of **neural and endocrine** adaptations known as the “stress response” or “stress cascade.”

Associated with changes in **glucocorticoids, catecholamines**, and much more.

Adrenal Glands: Anatomical and Functional Complexity



Long Description for Adrenal Glands: Anatomical and Functional Complexity

- This diagram illustrates the structure and hormone production functions of the adrenal gland, with emphasis on how it responds to stress.
- The adrenal gland is divided into two main parts:
 - **The cortex** (outer layer), which has three sub-layers:
 - **ZONA GLOMERULOSA: PRODUCES MINERALOCORTICOIDS SUCH AS ALDOSTERONE, CORTICOSTERONE, AND SMALL AMOUNTS OF PROGESTERONE.**
 - **ZONA FASCICULATA: PRODUCES GLUCOCORTICOIDS INCLUDING CORTISOL AND CORTISONE, WHICH ARE INVOLVED IN CHRONIC STRESS RESPONSES.**
 - **ZONA RETICULARIS: PRODUCES SEX HORMONES, SUCH AS DHEA, DHEA-S, ANDROGENS, AND SMALL AMOUNTS OF ESTROGEN.**
 - **The medulla** (inner part): Produces *catecholamines*, such as adrenaline and noradrenaline, which are involved in the acute stress response.
- Two orange ovals emphasize the role of adrenal hormones in stress:
 - **Acute Stress** triggers release of *catecholamines*.
 - **Chronic Stress** results in increased production of *cortisol*.

Functional Complexity of Cortisol

- Stimulates the liver to convert amino acids to glucose
- Stimulates increased glycogen in the liver
- Mobilizes fatty acids into the blood
- Increases coagulation
- Prevents the loss of sodium in urine
- Suppresses parts of the inflammatory response
- Maintains mood and emotional stability

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Spectrum of Adrenal Dysregulation

Addison's

Subclinical
Hypocortisolemia

Subclinical
Hypercortisolemia

Cushing's
Syndrome

Cortisol:
Very Low/Insufficient

Cortisol:
Low or Low-Normal

Cortisol:
Mildly Elevated

Cortisol:
Very High/Excessive

Variability of Cortisol Levels With Chronic Stress



Acute Stress: With acute stress, initial arousal and increased cortisol levels are reliable clinical findings.



Mid/Long-Term Stress: This adaptation phase often presents with elevated cortisol and decreased DHEA levels, *but laboratory and symptom presentations vary widely.*



Chronic Stress: May result in adrenal dysregulation with either *decreased or elevated cortisol* levels and usually decreased DHEA. Presentations may vary widely across individuals.

Acute Stress and the Catecholamine Response



Adaptive Effects of the Acute Stress Response

Sweating; cooling to improve exertion; reduced traction for predators.

Senses and perception sharpen. Time may appear to slow down.

Blood flow to muscles increases. Greater capacity for fight or flight responses.

Digestion stops. Blood is diverted from the gut to the arms and legs.

Hormones released that make the blood more coagulable to reduce blood loss if injured.

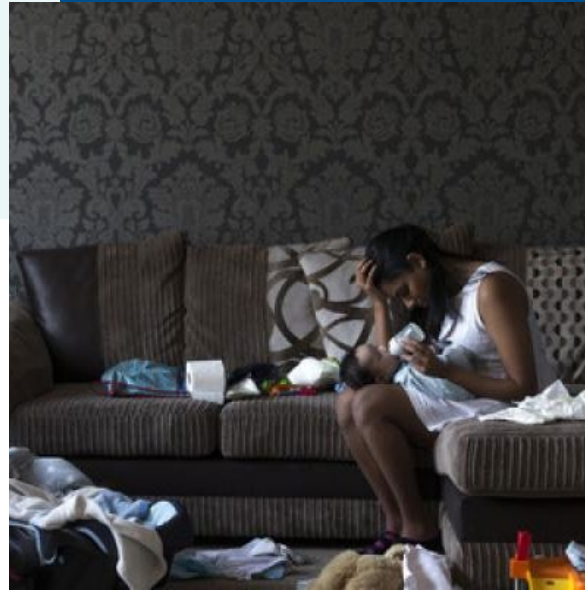


Tachycardia; increasing blood pressure and preparing for peak exertion.

Tachypnea; oxygenating the blood in preparation for fight or flight.

Chronic Stress and the Adrenal Response

A very different phenotype



Clinical Focus:

- Overview of allostatic load and adaptive allostasis.
- Role of the adrenal gland in response to environmental stress.



“Eustress”

- **Good stress, bad stress, and oxidative stress**
 - “Eustress” or manageable levels of life stress may enhance psychobiological resilience to oxidative damage.
 - Chronic stress promotes oxidative damage through frequent and sustained activation of the HPA axis.
- **Stress and cognition:**
 - Corticosteroid effects on cognition can turn from adaptive into maladaptive.

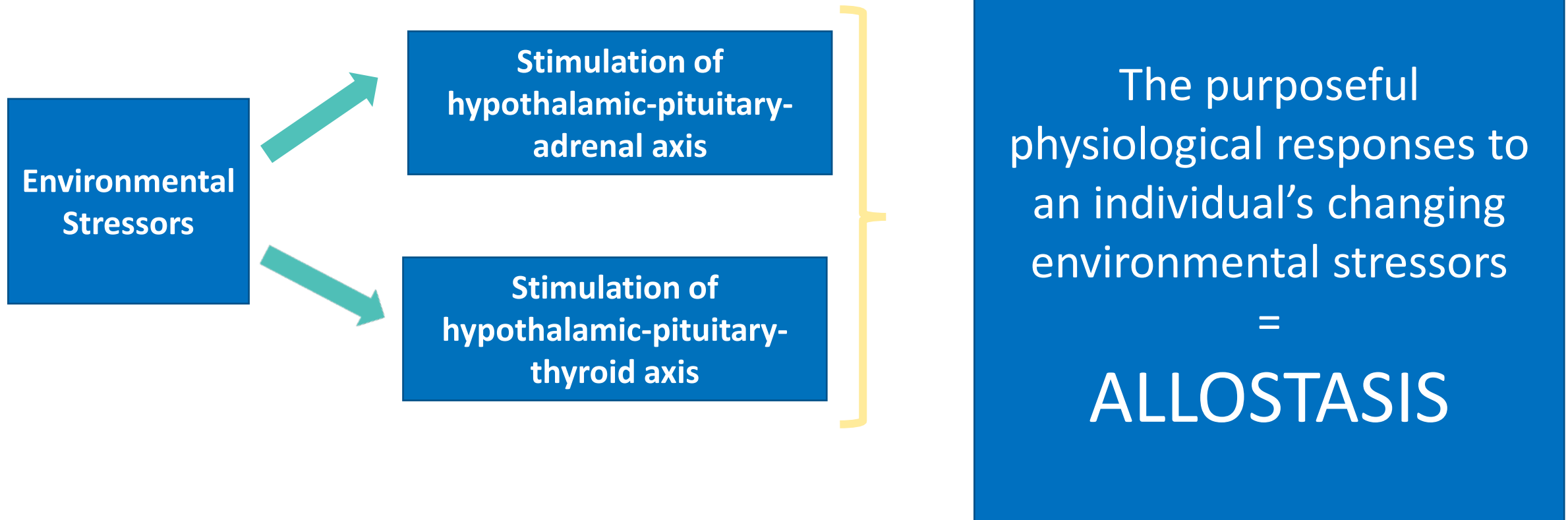
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“GOOD” VS “BAD” STRESS

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2. Aschbacher K, O'Donovan A, Wolkowitz OM, Dhabhar FS, Su Y, Epel E. Good stress, bad stress and oxidative stress: insights from anticipatory cortisol reactivity. *Psychoneuroendocrinology*. 2013;38(9):1698-1708. doi:10.1016/j.psyneuen.2013.02.004
3. de Kloet ER, Oitzl MS, Joëls M. Stress and cognition: are corticosteroids good or bad guys? *Trends Neurosci*. 1999;22(10):422-426. doi:10.1016/s0166-2236(99)01438-1

The HPAT Axis and Adaptive Allostasis

Environmental Stressors and Physiological Responses



Long Description for:

The HPAT Axis and Adaptive Allostasis

Environmental Stressors and Physiological Responses

- This graphic explains the concept of **allostasis**, which refers to the body's adaptive response to environmental stressors.
- On the left, a box labeled **Environmental Stressors** points to two physiological pathways:
 - **Stimulation of the hypothalamic-pituitary-adrenal (HPA) axis**
 - **Stimulation of the hypothalamic-pituitary-thyroid (HPT) axis**
- Both pathways represent how the body reacts to external challenges.
- On the right, a large box defines **Allostasis** as *“The purposeful physiological responses to an individual’s changing environmental stressors.”*
- In summary, allostasis is the body’s process of maintaining stability through change by activating hormonal systems that respond to stress.

Adaptive Allostasis

Environmental Stressors and Adrenal Response

Allostasis is positive and necessary to life.

It is different from homeostasis in that it is **what supports homeostasis.**

Allostasis allows for a **change in the set points** of the various physiological systems so that the body can respond adequately to environmental changes.

Adrenal hormones support these necessary **physiological adaptations** as the individual's environment and/or life circumstances change.

- Guidi J, Lucente M, Sonino N, Fava GA. Allostatic load and its impact on health: a systematic review. *Psychother Psychosom.* 2021;90(1):11-27. doi:10.1159/000510696
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Allostasis vs Allostatic Load

“Physiological Regulatory Demise”



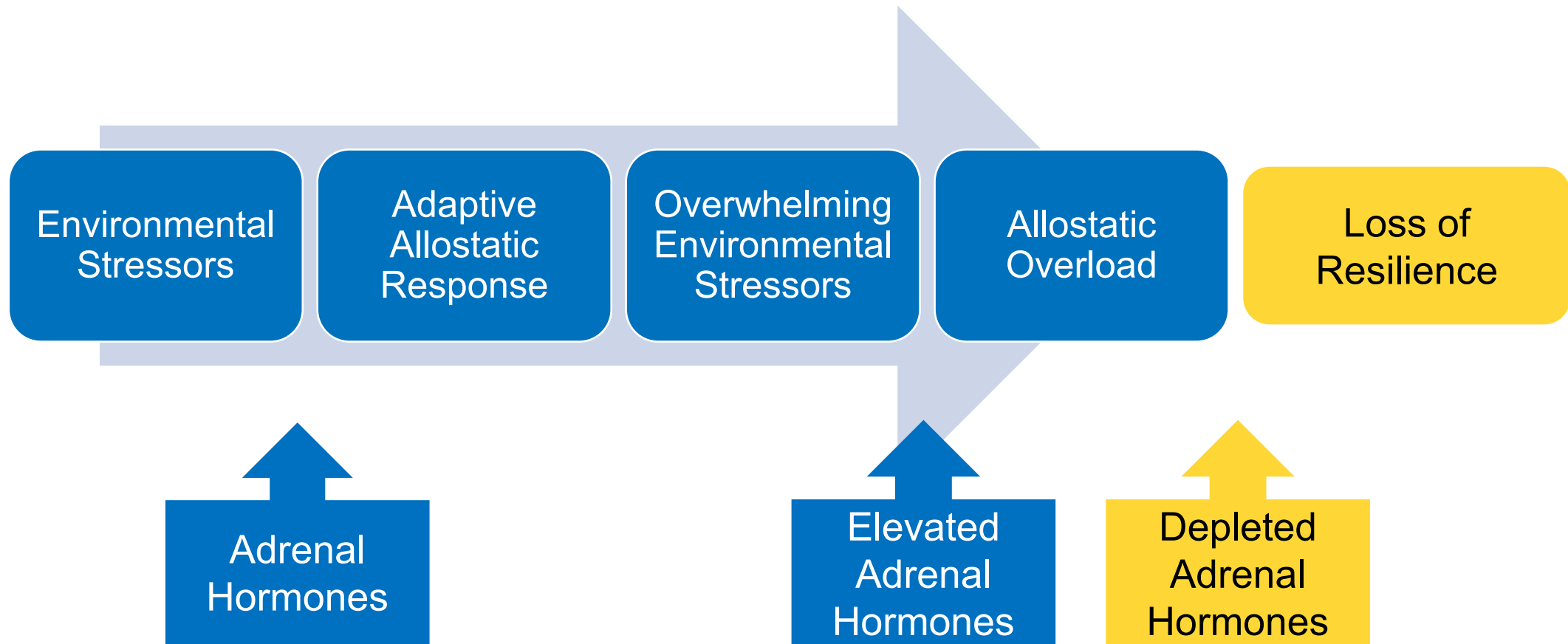
Allostatic load (AL) is the **wear and tear** that the body experiences due to the repeated use of allostatic responses and inefficient turning on or shutting off of these responses.



Allostatic load (AL) that is chronic and severe enough can result in the **loss of physiological resilience** that contributes to chronic disease.

Excessive Chronic Allostatic Load

Phases of the Dysregulatory Adrenal Response



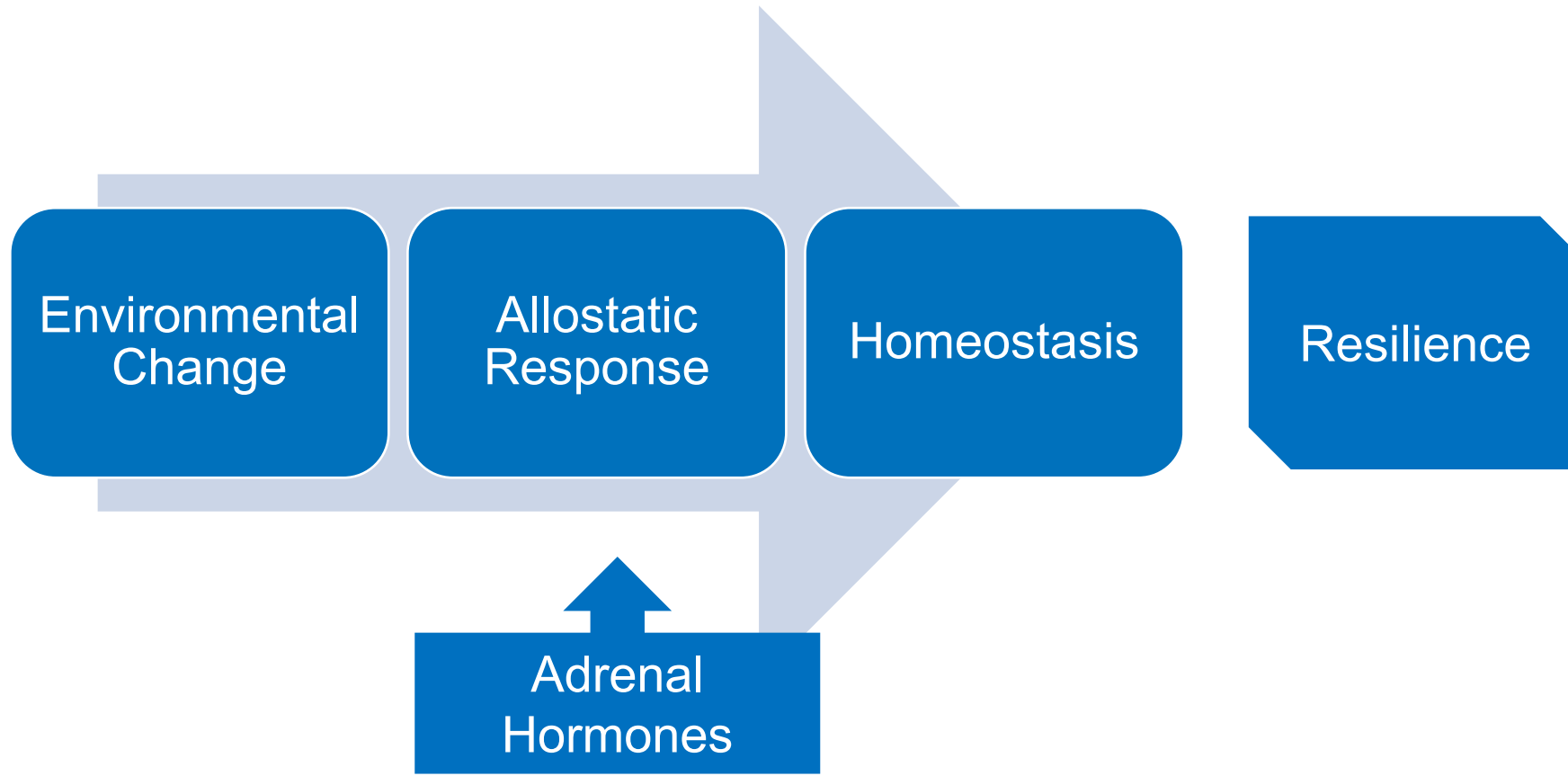
Long Description for: **Excessive Chronic Allostatic Load** Phases of the Dysregulatory Adrenal Response

This diagram explains the **phases of the dysregulated adrenal response** due to **excessive chronic allostatic load**, which can lead to reduced resilience over time.

- The process unfolds along a progression of stages as depicted with a large arrow and it is all tied to adrenal hormonal inputs
- It starts with Environmental Stressors, then Adaptive Allostatic Response, then Overwhelming Environmental Stressors, then Allostatic Overload
 - As one progresses further in these stages, adrenal hormones can become elevated
 - At the point of allostatic overload, one can develop adrenal hormone depletion
- The final stage in this progression is Loss of Resilience.
- The graphic shows a left-to-right progression from normal stress response to dysfunction, highlighting how chronic stress can impair hormonal health and overall resilience.

Allostasis and Resilience

The Role of Adrenal Hormones



Allostatic Load

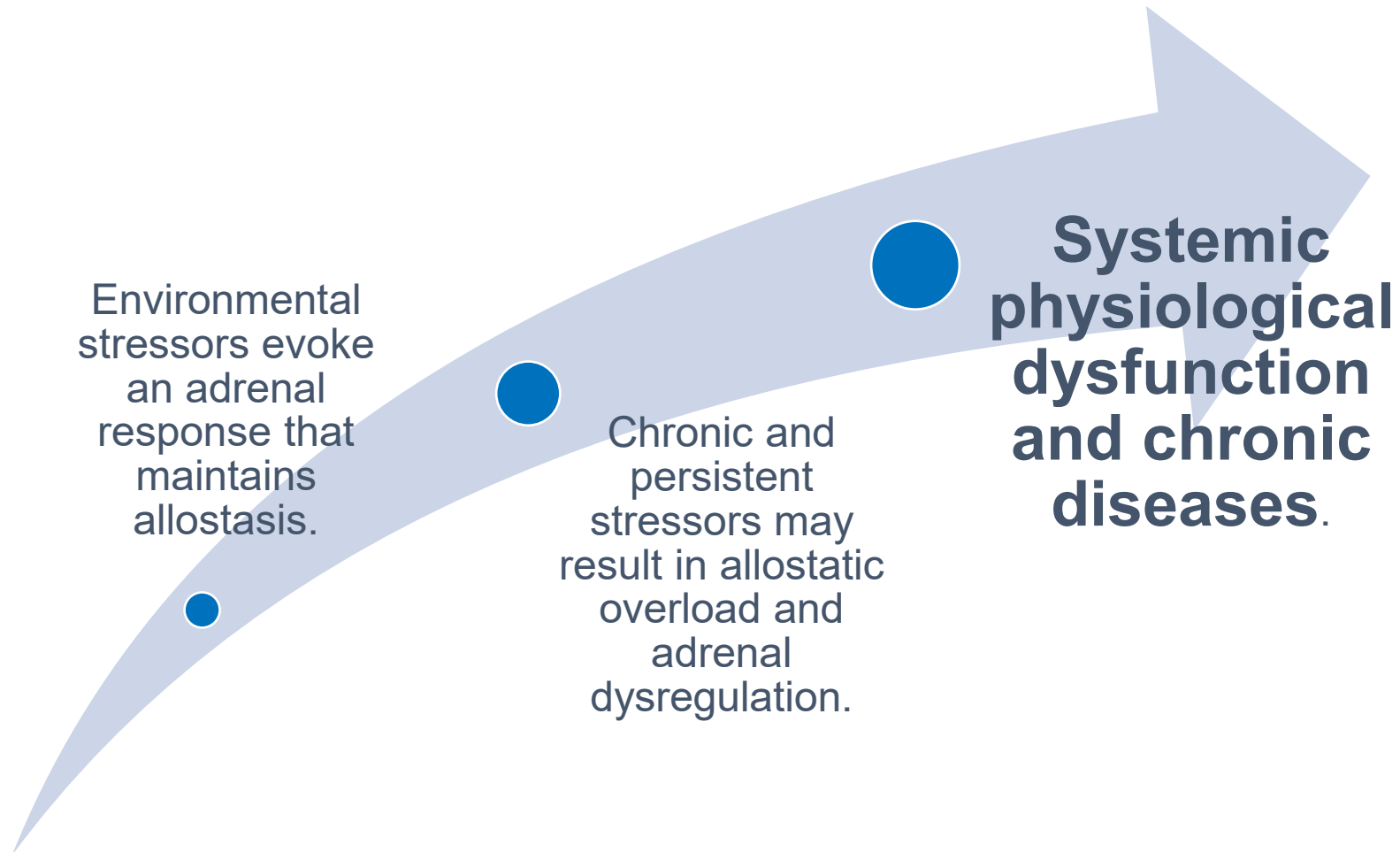


Long Description: Allostatic Load

- This image visually represents the concept of **allostatic load**, which is the cumulative strain on the body from repeated or chronic stress.
- On the left side, a hand stretches two rubber bands between two fingers, symbolizing **tension under pressure** — a metaphor for the body's adaptive response to stress.
- On the right side, four images show a rope progressively fraying:
 - A strong, intact rope symbolizes initial resilience.
 - Slight fraying indicates the beginning of wear from ongoing stress.
 - Increased fraying suggests weakening from prolonged load.
 - A fully broken rope represents **physiological breakdown or loss of resilience** due to excessive allostatic load.
- Together, the rubber band and fraying rope convey how continuous stress can overstretch the body's adaptive systems, ultimately leading to dysfunction if not addressed.

Chronic Stressors and Allostatic Overload

Adrenal Dysregulation and Chronic Disease



Long Description:

Chronic Stressors and Allostatic Overload

Adrenal Dysregulation and Chronic Disease

- This graphic explains the progression from chronic stress to chronic disease through the concept of **allostatic overload** and **adrenal dysregulation**.
- It follows an upward arrow across three stages:
 1. **Environmental stressors evoke an adrenal response that maintains allostasis**
 - This stage reflects the body's adaptive response to temporary stress.
 2. **Chronic and persistent stressors may result in allostatic overload and adrenal dysregulation**
 - Continued or intense stress begins to strain the adrenal system, reducing the body's ability to adapt.
 3. **Systemic physiological dysfunction and chronic diseases**
 - At this stage, prolonged dysregulation contributes to breakdown in multiple systems and the onset of chronic illness.
- The image shows that while allostasis helps maintain balance in the face of stress, sustained or excessive stress leads to overload, resulting in long-term physiological damage and disease risk.

Allostatic Load and Social Determinants of Health

In a study of 20,529 men with:

- High allostatic load
- Lower educational/socioeconomic status

Outcome:

- Greater than four-fold increased risk of cancer mortality



Manage the Multitasking



Multitasking = Stress

- Multitasking increases cortisol; combined with adrenaline overstimulating the brain = **mental fog or scrambled thinking**
- The prefrontal cortex = task management
- Has novelty bias = **attention easily hijacked**
- Multitasking = dopamine-addiction feedback loop, **rewarding the brain for losing focus & seeking external stimulation**

You Choose....



Is the glass half
empty?

OR

Is it half full?

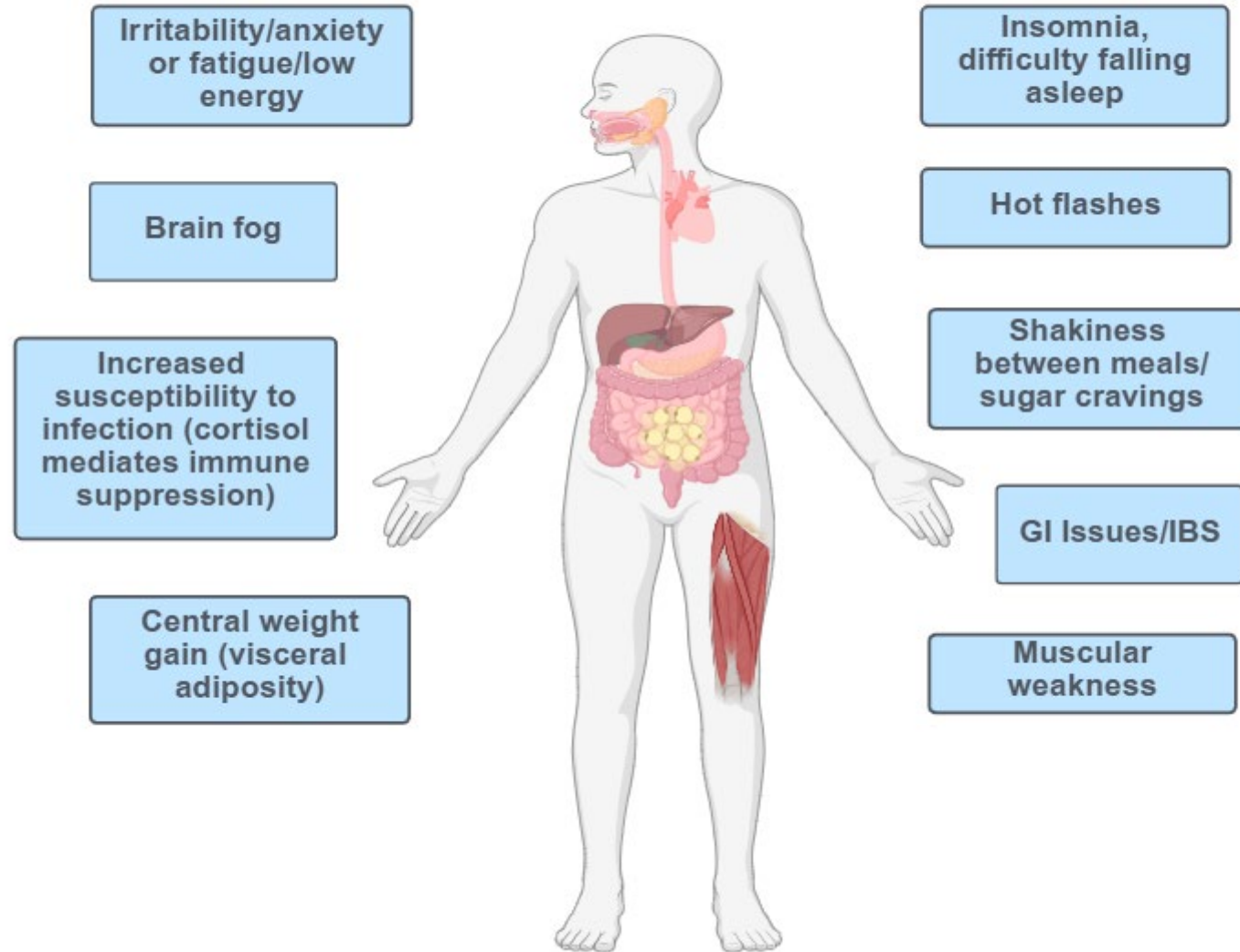
Perception
is the primary
driver of
physiologic
outcome.

Clinical Focus:

- Manifold clinical findings of adrenal dysregulation.



Chronically Elevated Cortisol: Associated Clinical Findings



Created with BioRender.com by the Institute for Functional Medicine

Long Description for Chronically Elevated Cortisol: Associated Clinical Findings

- This diagram highlights the clinical symptoms associated with **chronically elevated cortisol** levels, using a human body illustration with labeled callouts.
- Symptoms listed include:
 - Irritability, anxiety, or fatigue/low energy
 - Brain fog
 - Increased susceptibility to infection (due to cortisol's immune-suppressing effects)
 - Central weight gain, particularly visceral adiposity
 - Insomnia or difficulty falling asleep
 - Hot flashes
 - Shakiness between meals or sugar cravings
 - Gastrointestinal issues or I B S (irritable bowel syndrome)
 - Muscular weakness
- These symptoms reflect the widespread effects of cortisol dysregulation on the nervous system, metabolism, immune response, digestion, and sleep. The graphic emphasizes that chronic stress and prolonged cortisol elevation can contribute to a wide range of physiological dysfunctions.

Clinical Findings: Decreased Cortisol Levels

MENTAL HEALTH CONDITIONS AND ENIGMATIC SYNDROMES

- Atypical depression
- Seasonal affective disorder (SAD)
- Postpartum depression
- Panic attacks/generalized anxiety disorder
- Bipolar II disorder
- PTSD
- CFIDS
- Fibromyalgia

Clinical Findings: Increased Cortisol Levels

Matrix Node	Condition
Transport	Type 2 Diabetes
Defense and Repair	Immune Suppression
Communication	Hypothyroid, Weight Gain, Central Obesity
Energy	Melancholic Depression, Anorexia Nervosa, Panic Disorders, Obsessive Compulsive Disorders, Memory Impairment
Structural Integrity	Osteoporosis
Assimilation	Malnutrition, GI Dysfunction
Biotransformation and Elimination	Adult-Onset Acne
Modifiable Lifestyle Factors	Chronic Excessive Exercise

Clinical Findings: Decreased Cortisol Levels

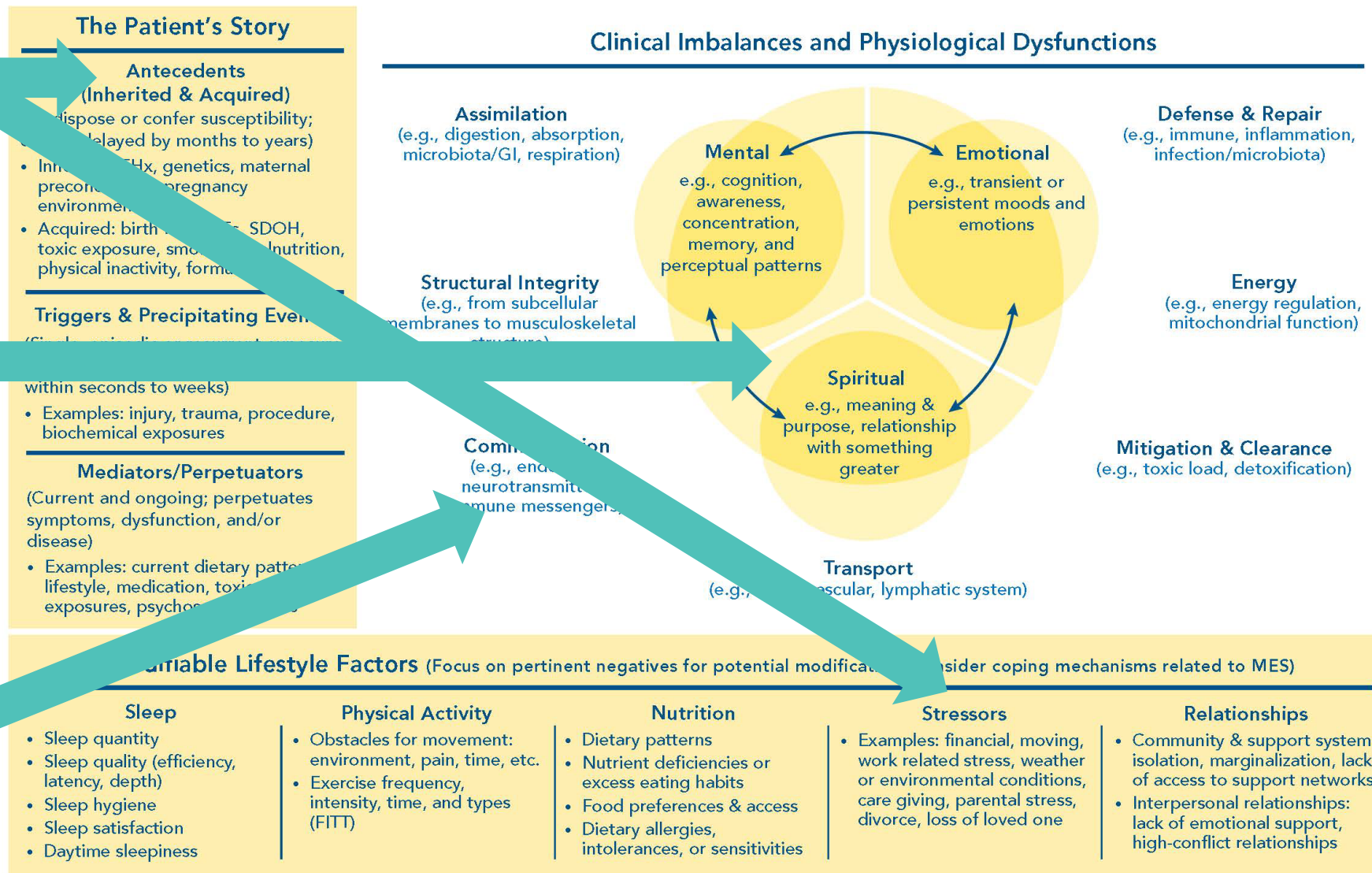
Matrix Node	Signs and Symptoms
Transport	• Low blood pressure and/or drop of blood pressure on standing • Dizziness on standing or bending • Fibromyalgia (musculoskeletal tender points) • Easy bruising/slow healing of cuts
Defense and Repair	• Increased susceptibility to inflammation and allergies (not infections)
Communication	• Early-onset perimenopause or menopause • Craving salty foods, pickles, etc. • “Low blood sugar” symptoms: dizzy, irritable, symptoms relieved by food • Unstable body temperatures
Energy	• Atypical depression • Postpartum depression • Seasonal affective disorder (SAD) • Panic attacks/generalized anxiety disorder • Bipolar II disorder • Post-traumatic stress syndrome (PTSD) • Chronic fatigue and immune dysfunction syndrome (CFIDS) • Fatigue, apathy, unmotivated • Absent-minded/poor concentration • Increased/excessive sleep but poor quality

Functional Medicine Matrix (Teaching)

What stressors
has the patient
experienced?

How did they
perceive those
situations?

What was their
adaptive
allostatic
response?



Long Description for questions designed to elicit specific matrix information

- Reflective clinical questions are connected to certain areas of the Functional Medicine Matrix.
- Large arrows from each question guide the viewer to ask these key questions when reviewing a patient's case:
 - **What stressors has the patient experienced?**
This question points to both the Antecedents portion of the “Patient’s Story” section of the matrix and also to Stressors in the Modifiable Lifestyle factors section
 - **How did they perceive those situations?**
This guides attention to the center of the matrix, focusing on **mental, emotional, and spiritual** dimensions
 - **What was their adaptive allostatic response?**
This points to the Communication node of the matrix

Clinical Focus:

- Antecedents, triggers, and mediators of adrenal dysregulation.



Antecedents, Triggers, and Mediators: Environmental Stressors



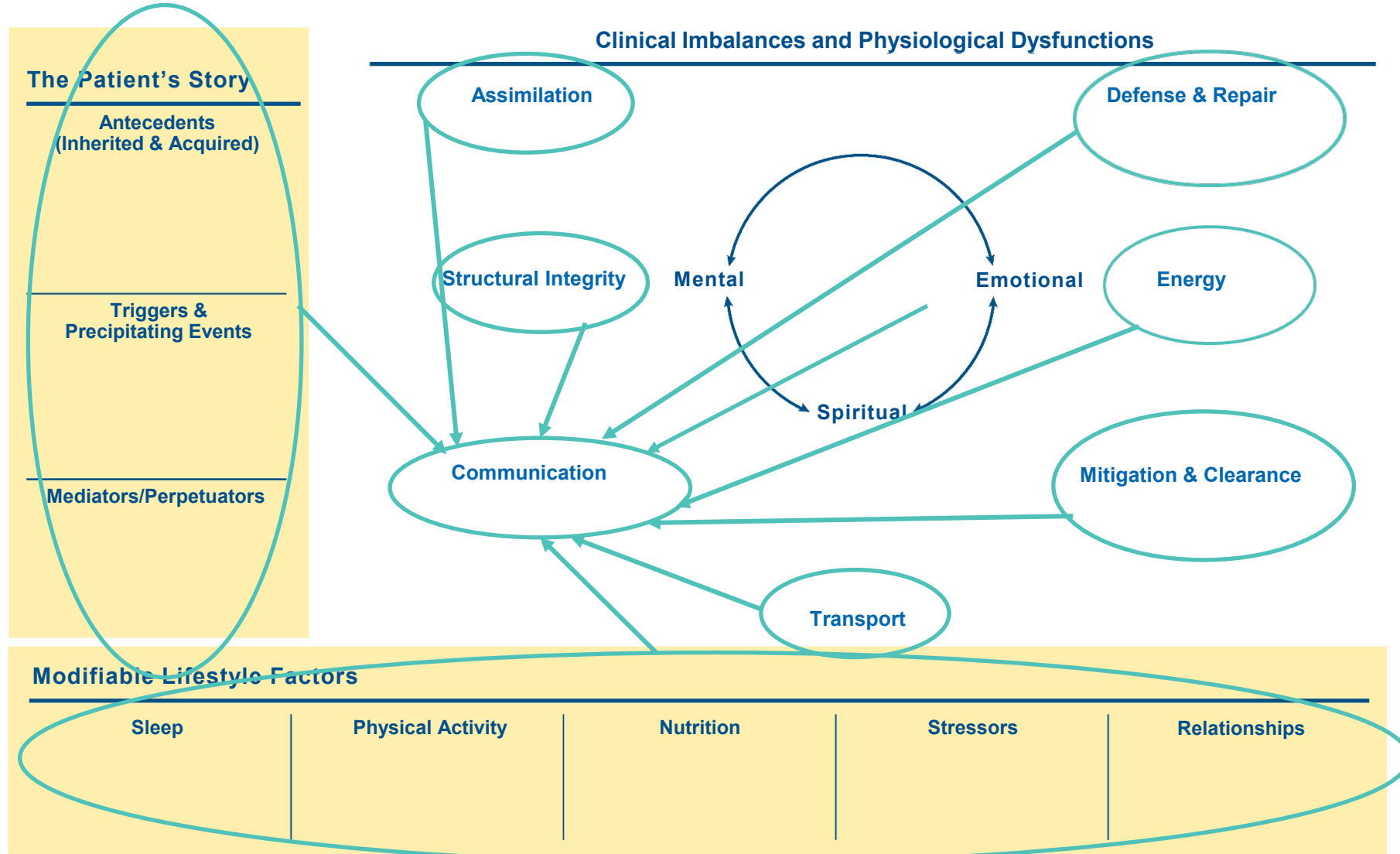
Any type of environmental stressor, external or internal, physiological or psychological, has the capacity to chronically elevate and maintain the stress response, causing adrenal dysregulation and leading to the development of chronic disease.



These environmental stressors represent the antecedents, triggers, and mediators of adrenal dysregulation.

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Functional Medicine Matrix



External and Internal Environmental Stressors That Contribute to Adrenal Dysregulation

Common ATMs of Adrenal Dysregulation

Genetics

Acute and
Chronic
Stressors

Inflammation,
Infection,
Trauma, Toxins

Dysbiosis
Intestinal
Permeability

Structural
Imbalances,
Pain

Dysinsulinemia
Adiposity
Obesity

Impaired
Biotransformation
and Methylation

Food Allergy,
Sensitivity,
and/or
Intolerance

Poor Diet
Nutrient
Deficiencies

Poor Sleep
Overexercise

Alcohol
Tobacco
Drugs

Hostile
Relationships

ATMs of Adrenal Dysregulation: STAINS Mnemonic

STRESS	TOXINS	ALLERGENS/ ADVERSE FOOD REACTIONS	INFECTIONS/ INFLAMMATION/ INTESTINES	NUTRITION	SLEEP SEDENTARY SDOH
- Physical - Mental	- Biologic, elemental, synthetic - Impaired detoxification	- Food - Mold - Dust - Animal products - Pollens - Chemicals	- Bacteria - Yeast - Parasites - Viruses - Gut - Dysfunctions (DIGIN)	- Nutrient insufficiencies - Nutrient excesses	Various mechanisms

Communication and Hormone Balance: Part 2

Addressing Adrenal Dysregulation and Chronic Stress

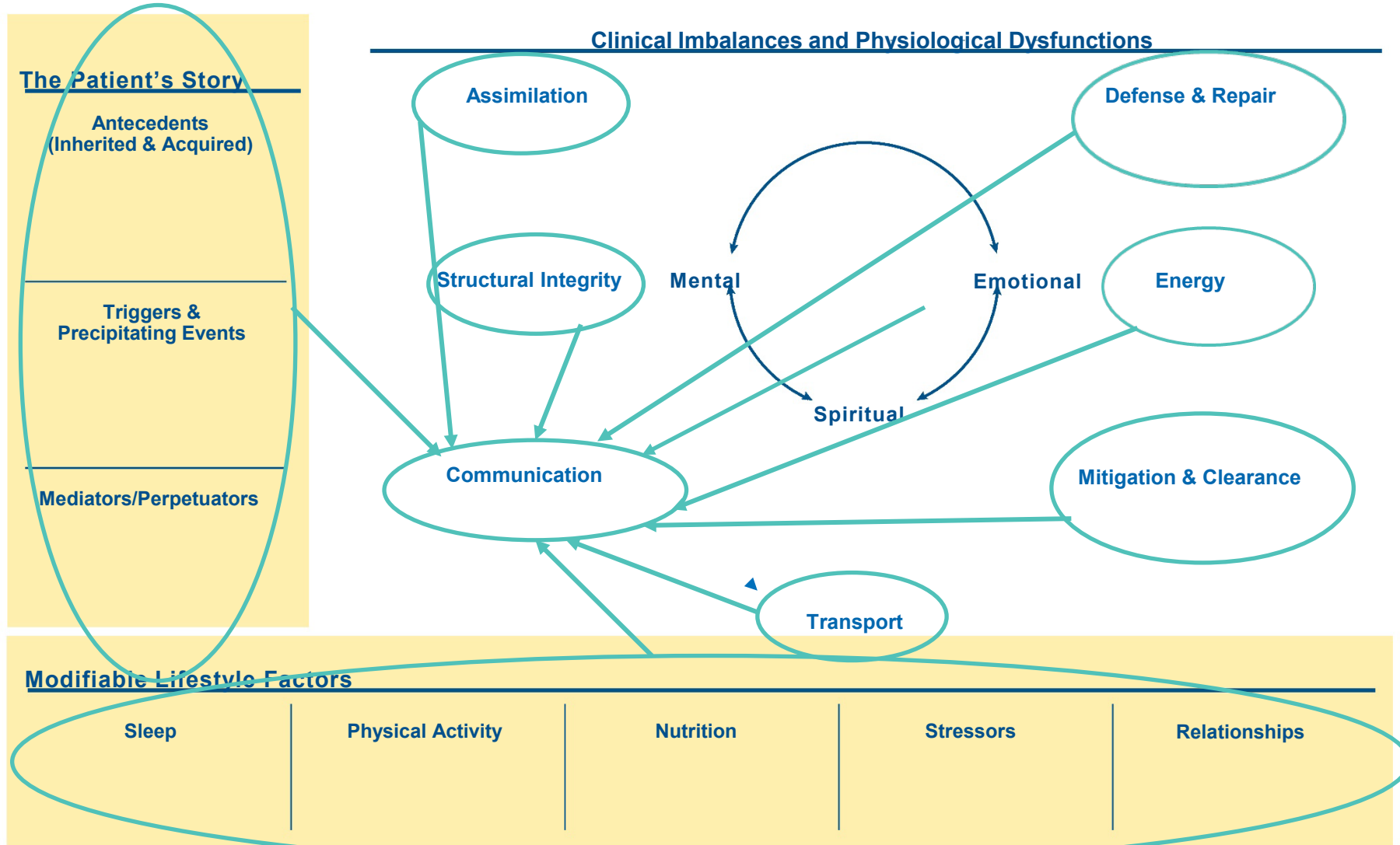
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus:

- Pathophysiological mechanisms of adrenal dysregulation mediated by matrix node dysfunctions and lifestyle behaviors.



Functional Medicine Matrix



Mechanisms of Matrix Node Imbalances and Modifiable Lifestyle Factors as Mediators of Adrenal Dysregulation

Assimilation Node

Maldigestion and Malabsorption



Maldigestion and malabsorption from any cause can reduce the bioavailability of micronutrients necessary for optimal adrenal function such as B-complex vitamins, vitamin C, and magnesium.

See: "References: Assimilation Node"

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Structural Integrity Node

Intestinal Permeability: Mechanisms of Adrenal Dysregulation



Compromised gut barrier integrity can allow inflammatory cytokines and endotoxins such as **LPS** to enter the bloodstream and trigger systemic inflammation. Inflammation can in turn trigger the HPA axis, leading to adrenal dysregulation and imbalances in cortisol levels.



Gut permeability is linked to altered **serotonin** and **GABA** production, both of which regulate the HPA axis and mood. Low serotonin can impair cortisol regulation.

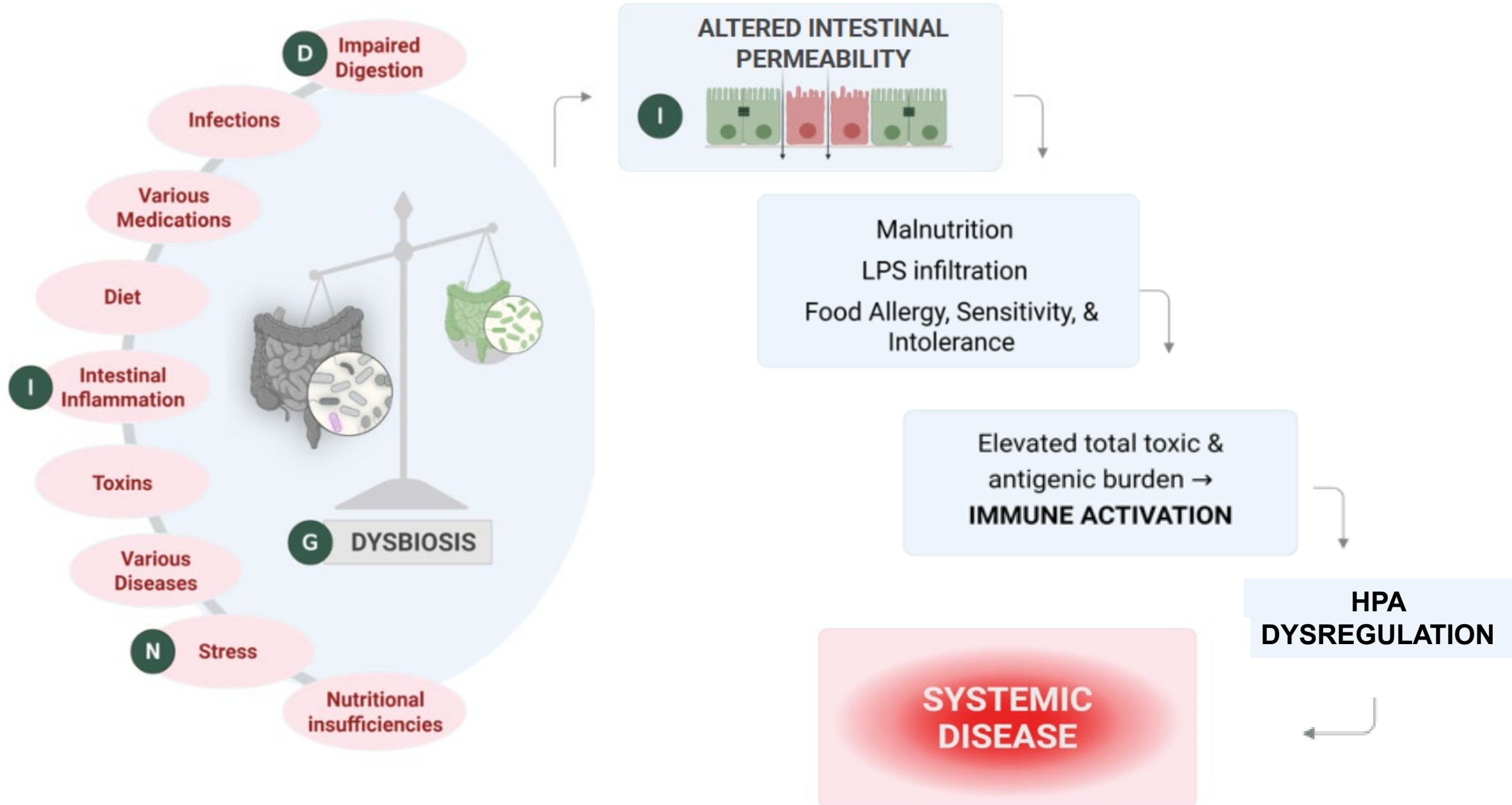


Increased gut permeability can impact **insulin sensitivity**, leading to blood sugar fluctuations that also impact HPA axis.

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STRUCTURAL INTEGRITY

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Long Description for graphic depicting how dysbiosis can lead to systemic disease

- This graphic illustrates how gut dysbiosis and DIGIN dysfunctions can lead to systemic disease through a cascade of physiological disruptions.
- **Left Side: Contributors to Dysbiosis**
 - A semi-circle on the left shows factors contributing to **dysbiosis**, which refers to an imbalance in the gut microbiota:
- **Right Side: Consequences of Dysbiosis**
 - The graphic flows through a sequential process showing how dysbiosis leads to broader dysfunction:
 - First, **Altered intestinal permeability**
 - This leads to **malnutrition, L P S (lipopolysaccharide) infiltration**, and **food allergies or sensitivities**.
 - These exposures increase the **total toxic and antigenic burden**, triggering **immune activation**.
 - Immune activation contributes to **H P A axis dysregulation** (hypothalamic-pituitary-adrenal axis).
- Ultimately, these cascades culminate in **systemic disease**, shown in a red glowing box.

References

DIGIN DYSFUNCTIONS

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Defense and Repair Node

Systemic Inflammation: Mechanisms of Adrenal Dysregulation

Cytokines can trigger HPA-axis activation and development of glucocorticoid excess.

In prolonged disease, a **glucocorticoid resistance** develops, exhibiting normal or decreased ACTH and elevated cortisol levels.

Energy Node

Mitochondrial Dysfunction: Mechanisms of Adrenal Dysregulation

Mitochondria are the site of synthesis for adrenal hormones.



Energy Node

Clinical Relevance and Key Clinical Takeaway

Clinical findings of mitochondrial dysfunction overlap with those of adrenal insufficiency:

- Chronic fatigue, fatigue, burnout and poor resilience to stress
- Hypotension
- Cognitive impairment (brain fog)
- Difficulty recovering from stress

References:

ENERGY NODE -MITOCHONDRIAL DYSFUNCTION: MECHANISMS OF ADRENAL DYSREGULATION

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Mitigation and Clearance Node

Total Toxin Load: Mechanisms of Adrenal Dysregulation

Pesticides, plasticizers, dioxins, PCBs, and polycyclic aromatic hydrocarbons can **accumulate** in the adrenal gland and cause **lesions**.

Pesticides and polycyclic aromatic hydrocarbons can damage the adrenal gland by **bioactivation of CYP450 enzymes**.

BPA may affect the activity as well as the expression of **enzymes** that are involved with **steroidogenesis**.

Microplastics can interfere with various hormone **receptors** and disrupt the hypothalamic-pituitary-adrenal (HPA) axis.

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MITIGATION AND CLEARANCE NODE – TOTAL TOXIN LOAD: MECHANISMS OF ADRENAL DYSREGULATION

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Transport/Communication Node:

Type 2 Diabetes Mellitus: Mechanisms of Adrenal Dysregulation

Hyperglycemia in type 2 diabetes mellitus (T2DM) is associated with HPA axis dysregulation.

This dysregulation is attributed to **disruption of the negative feedback mechanism** that involves ACTH and cortisol.

Modifiable Lifestyle Factors:

Mechanisms of MLFs as Mediators of Adrenal Dysregulation



Modifiable Lifestyle Factors

Sleep Deprivation: Mechanisms of Adrenal Dysregulation

Sleep deprivation (max of 4 hrs. sleep/1-6 consecutive days) is associated with dysregulated cortisol secretion patterns the next day:

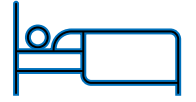
Reduced morning
cortisol awakening
response (CAR)

Increased
afternoon/evening
cortisol levels

Reduced
reactivity/slower
recovery of the
cortisol response

Modifiable Lifestyle Factors

Sleep Quality: Mechanisms of Adrenal Dysregulation



Studies on the association between sleep quality and HPA stress responsiveness have yielded mixed results.

- Insomnia is associated with overall increase of ACTH and cortisol secretion.
- Sleep deprivation is associated with decreased ACTH response to stress.
- Other studies do not show these associations.



References

MLFS – SLEEP QUALITY: MECHANISMS OF ADRENAL DYSREGULATION

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Modifiable Lifestyle Factors

Diet Quality: Mechanisms of Adrenal Dysregulation



Diets high in refined sugar have a complex impact on adrenal function and cortisol balance.

Short term: Sugary “comfort foods” can provide momentary, short-term relief from stress by diminishing acute cortisol release, temporarily suppressing HPA axis reactivity.

Long term: Chronic overconsumption of high-sugar diets can dysregulate HPA function. Diets high in added sugars are linked to metabolic disturbances that affect cortisol regulation over time.

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Modifiable Lifestyle Factors

Nutrient Deficiencies: Mechanisms of Adrenal Dysregulation



Vitamin C is needed in organs that synthesize catecholamines (epinephrine, norepinephrine, dopamine) such as in the brain and adrenal glands.



Essential fatty acid deficiency is linked to low plasma cortisol.



Zinc levels are reduced after periods of stress, when adrenalin levels go up.



Pantothenic acid deficiency causes impaired adrenocortical function and abnormal stress responses.



Magnesium deficiency causes elevated HPA set-point ($\uparrow$ CRH, $\uparrow$ ACTH); stress increases requirement for magnesium.

References

NUTRITIONAL INSUFFICIENCIES

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Modifiable Lifestyle Factors

Quality of Relationships: Mechanisms of Adrenal Dysregulation

- Loneliness, also termed perceived social isolation (PSI), is a form of chronic stress that has been linked to **disruptions in the HPA axis** that can lead to changes in cortisol production and regulation.
- Physiologically, the body interprets chronic psychological stressors such as **loneliness as a threat**, activating the hypothalamic-pituitary-adrenocortical (HPA) axis and increasing cortisol. This can lead to allostatic overload.



Modifiable Lifestyle Factors

Quality of Emotions: Mechanisms of Adrenal Dysregulation

Stress elicits both emotional and hormonal qualities that are regulated by the HPA axis. Studies have demonstrated:



Momentary negative emotions are associated with elevated cortisol levels.



Momentary positive emotions are associated with decreased cortisol levels.

You Choose....



← Is the glass half empty?

OR

← Is it half full?

**Perception
is the
primary
driver of
physiologic
outcome.**

Clinical Focus:

- Pathophysiological mechanisms of adrenal dysregulation as a mediator of other matrix node dysfunctions.



Functional Medicine Matrix

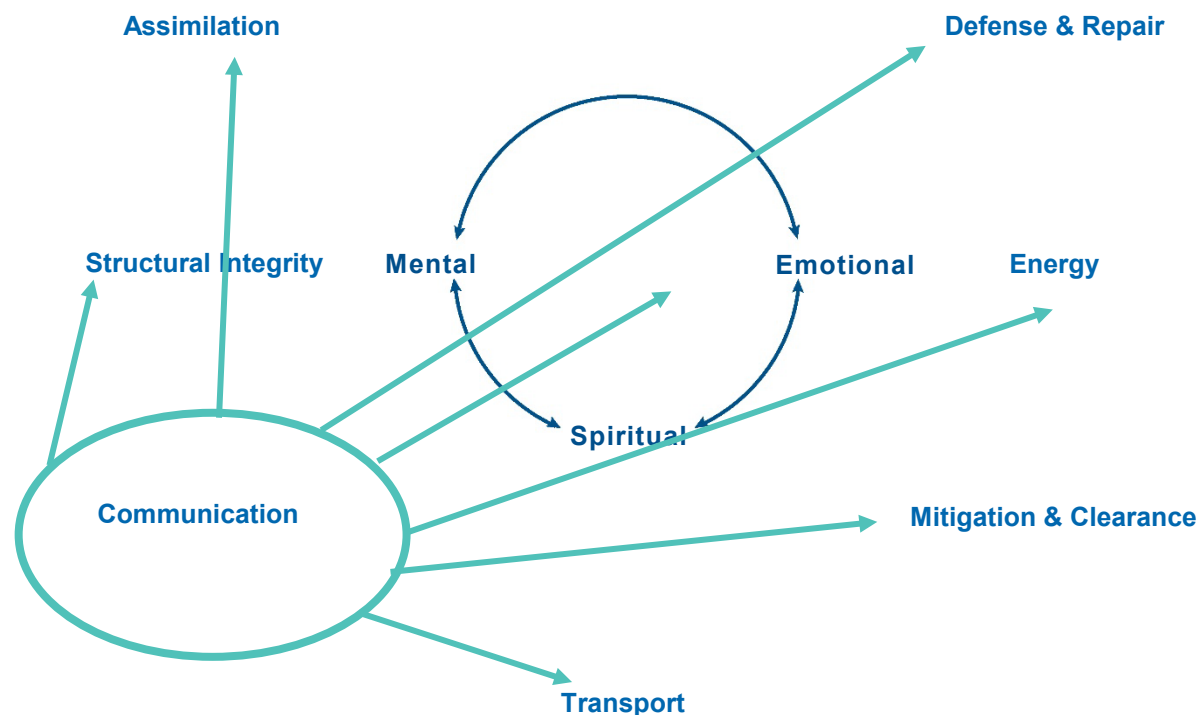
The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions



Systemic Effects
of Adrenal
Dysregulation and
the Chronic
Stress Response

Modifiable Lifestyle Factors

Sleep

Physical Activity

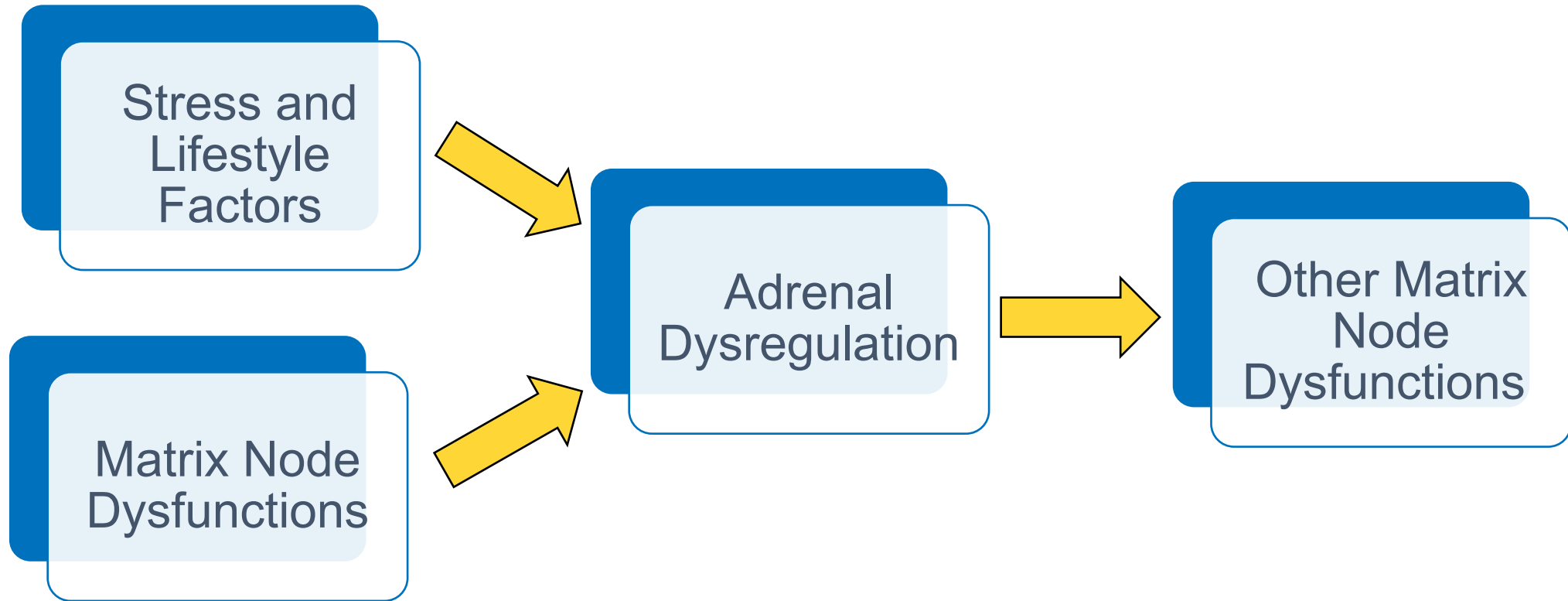
Nutrition

Stressors

Relationships

Complexity of Causative Relationships

Adrenal Dysregulation, Stress, Lifestyle, and Matrix Node Functions



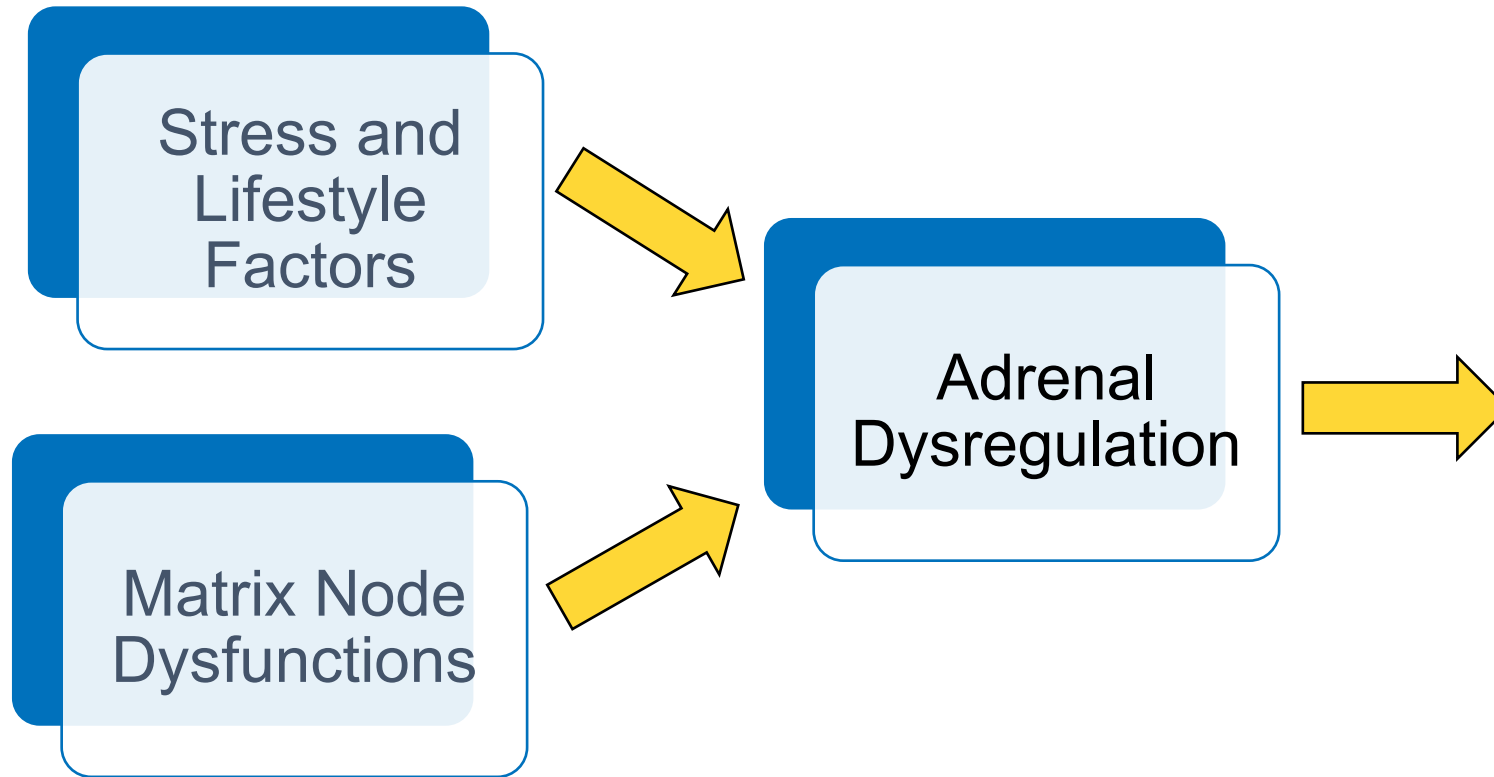
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Long Description: Complexity of Causative Relationships

- A flowchart illustrating how **adrenal dysregulation** can both result from and contribute to dysfunction in other matrix nodes
- Two sources lead to **adrenal dysregulation**:
 - **Stress and lifestyle factors**
 - **Matrix node dysfunctions**
- In turn, **adrenal dysregulation** can cause or worsen **other matrix node dysfunctions**, creating a feedback loop of physiological imbalance.

Complexity of Causative Relationships

Adrenal Dysregulation, Stress, Lifestyle, and Matrix Node Functions



- Allen JO, Mezuk B, Byrd DR, et al. Mechanisms of cardiometabolic health outcomes and disparities: what characteristics of chronic stressors are linked to HPA-axis dysregulation? *J Aging Health*. 2022;34(3):448-459. doi:10.1177/08982643221085903
- Le-Ha C, Herbison CE, Beilin LJ, et al. Hypothalamic-pituitary-adrenal axis activity under resting conditions and cardiovascular risk factors in adolescents. *Psychoneuroendocrinology*. 2016;66:118-124. doi:10.1016/j.psyneuen.2016.01.002

Transport Node Dysfunction:

Adrenal Dysregulation as Mediator

HPA axis dysregulation has been associated with cardiometabolic health outcomes such as:

Dyslipidemia

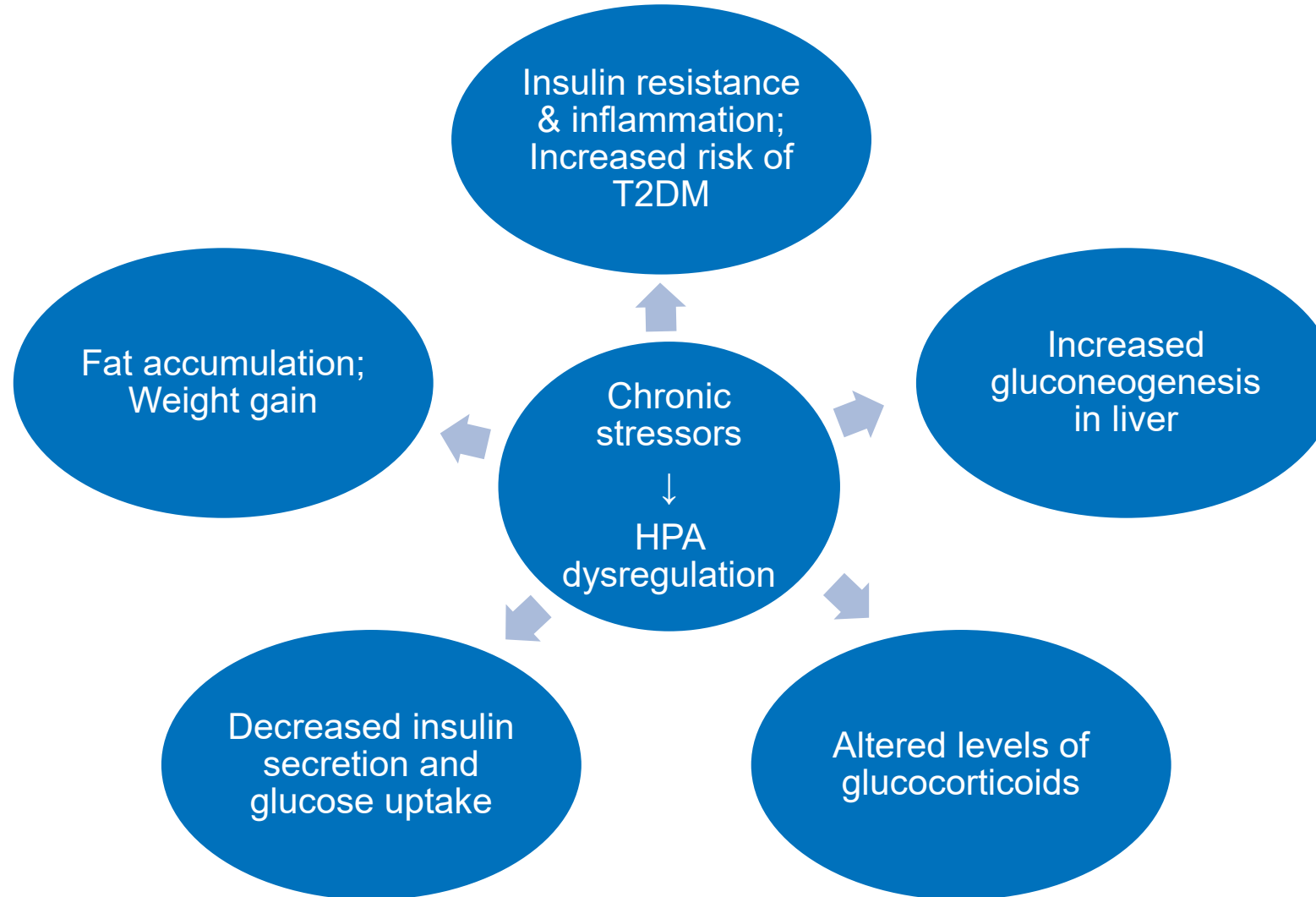
Hypertension

Insulin
Resistance

Central
Adiposity

- Allen JO, Mezuk B, Byrd DR, et al. Mechanisms of cardiometabolic health outcomes and disparities: what characteristics of chronic stressors are linked to HPA-axis dysregulation? *J Aging Health*. 2022;34(3):448-459. doi:10.1177/08982643221085903
- Le-Ha C, Herbison CE, Beilin LJ, et al. Hypothalamic-pituitary-adrenal axis activity under resting conditions and cardiovascular risk factors in adolescents. *Psychoneuroendocrinology*. 2016;66:118-124. doi:10.1016/j.psyneuen.2016.01.002

Transport Node Dysfunction: Mediators Induced by Adrenal Dysregulation



Long Description:

Transport Node Dysfunction:

Mediators Induced by Adrenal Dysregulation

- The graphic illustrates how chronic stress and hypothalamic-pituitary-adrenal (HPA) axis dysregulation contribute to transport node dysfunction
- At the center is “Chronic stressors lead to H P A dysregulation.”
- From this H P A dysregulation, the mediators induced are as follows:
 - Insulin resistance & inflammation; Increased risk of Type 2 Diabetes Mellitus (T 2 D M)
 - Increased gluconeogenesis in liver
 - Altered levels of glucocorticoids
 - Decreased insulin secretion and glucose uptake
 - Fat accumulation; Weight gain

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PATHOPHYSIOLOGICAL MECHANISMS OF ADRENAL DYSREGULATION: TRANSPORT NODE – INSULIN METABOLISM

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- Sharma VK, Singh TG. Chronic stress and diabetes mellitus: interwoven pathologies. *Curr Diabetes Rev.* 2020;16(6):546-556. doi:10.2174/1573399815666191111152248

Structural Integrity Node Dysfunction (Visceral Adiposity): Adrenal Dysregulation as Mediator

HPA axis dysregulation and increased cortisol are associated with **accumulation of fat cells**, weight gain, and increased risk for central obesity.

Increased glucocorticoid levels stimulate **relocation of fat storage** from peripheral subcutaneous adipose tissue to visceral adipose tissue.

- Incollingo Rodriguez AC, Epel ES, White ML, Standen EC, Seckl JR, Tomiyama AJ. Hypothalamic-pituitary-adrenal axis dysregulation and cortisol activity in obesity: a systematic review. *Psychoneuroendocrinology*. 2015;62:301-318. doi:10.1016/j.psyneuen.2015.08.014
- Lengton R, Schoenmakers M, Penninx BWJH, Boon MR, van Rossum EFC. Glucocorticoids and HPA axis regulation in the stress-obesity connection: a comprehensive overview of biological, physiological and behavioural dimensions. *Clin Obes*. 2025;15(2):e12725. doi:10.1111/cob.12725

Assimilation Node Dysfunction (DIGIN Factors): Adrenal Dysregulation as Mediator

Various cells of the gut such as epithelial cells, immune cells, and enteroendocrine cells express **cortisol receptors** and can be directly affected by HPA dysregulation and increased levels of cortisol. Effects may include:



Alterations in gut microbial composition.



Increased gut permeability: microbes, microbial metabolites, cytokines, and other immune molecules can enter circulation, contributing to inflammation.



Changes in gut transit time.



Altered metabolism and absorption of nutrients.

Mitigation and Clearance Node Dysfunction (CYP Enzymes): Adrenal Dysregulation as Mediator

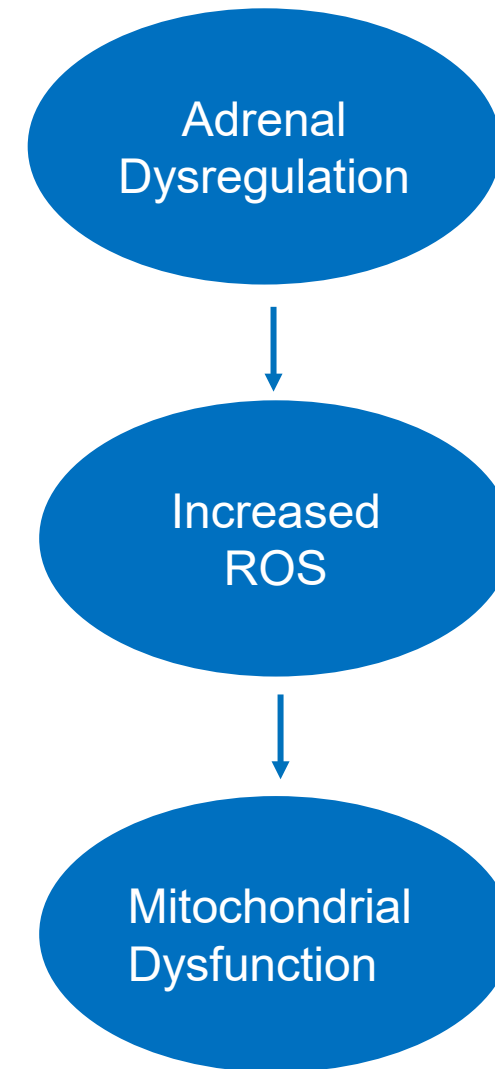
Elevated glucocorticoid concentrations **stimulate** the synthesis and expression of certain **CYP enzymes**.

Dysregulation of HPA axis feedback and increased plasma cortisol levels can affect CYP enzymes, affecting drug efficacy and toxicity.

- Tseilikman V, Dremencov E, Tseilikman O, Pavlovicova M, Lacinova L, Jezova D. Role of glucocorticoid- and monoamine-metabolizing enzymes in stress-related psychopathological processes. *Stress*. 2020;23(1):1-12. doi:10.1080/10253890.2019.1641080
- Konstandi M, Johnson EO. Age-related modifications in CYP-dependent drug metabolism: role of stress. *Front Endocrinol (Lausanne)*. 2023;14:1143835. doi:10.3389/fendo.2023.1143835

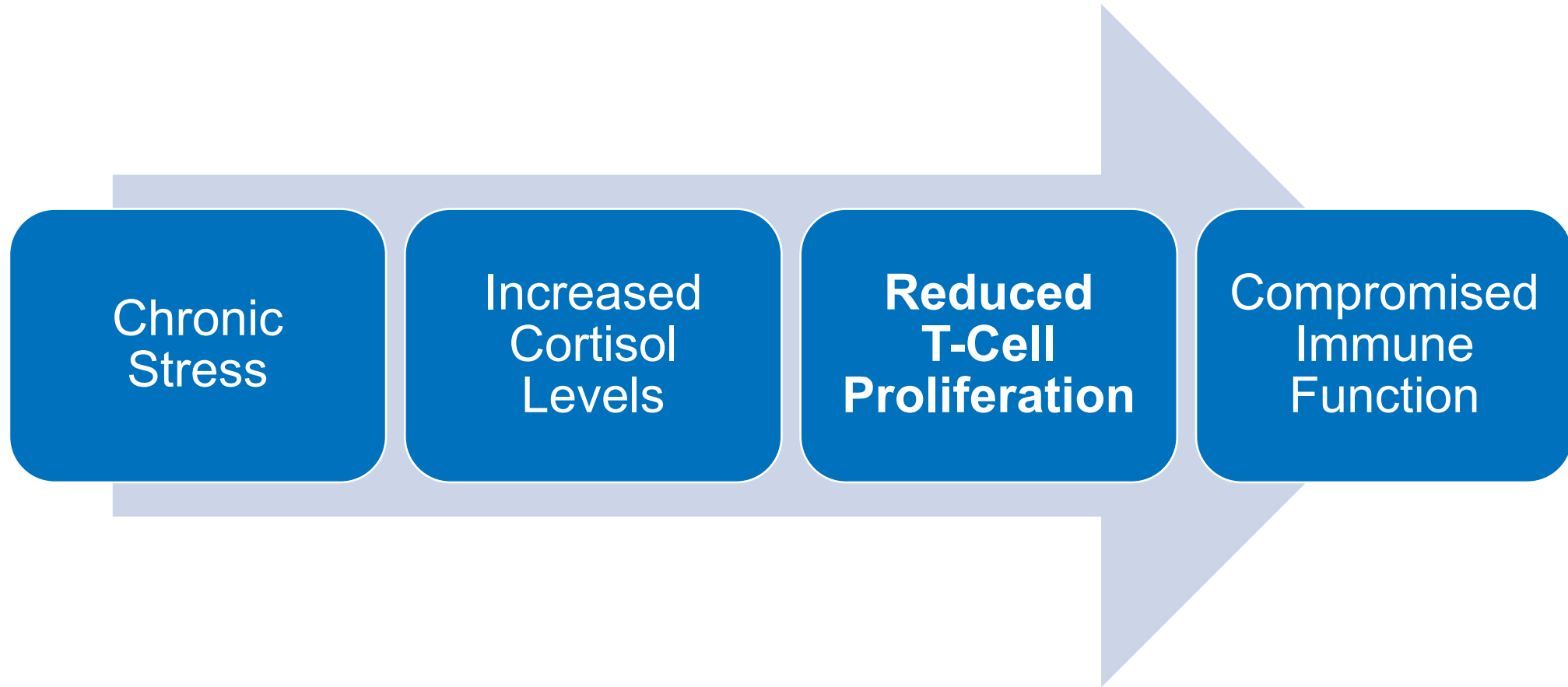
Energy Node Dysfunction (Mitochondria): Adrenal Dysregulation as Mediator

Dysregulation of adrenal hormones can contribute to **increased ROS**, a known mediator of mitochondrial dysfunction.



- Patani A, Balram D, Yadav VK, Lian KY, Patel A, Sahoo DK. Harnessing the power of nutritional antioxidants against adrenal hormone imbalance-associated oxidative stress. *Front Endocrinol (Lausanne)*. 2023;14:1271521. doi:10.3389/fendo.2023.1271521
- Picard M, McEwen BS, Epel ES, Sandi C. An energetic view of stress: focus on mitochondria. *Front Neuroendocrinol*. 2018;49:72-85. doi:10.1016/j.yfrne.2018.01.001

Defense and Repair Node Dysfunction (Immune System): Adrenal Dysregulation as Mediator



Matrix Node Imbalances and Mechanisms of Adrenal Dysregulation

Transport	Hyperglycemia in type 2 diabetes mellitus (T2DM) is associated with (HPA) axis dysregulation.
Structural Integrity	Compromised gut barrier integrity can lead to systemic inflammation and altered serotonin production and can impact insulin sensitivity; all can lead to HPA-axis dysregulation.
Assimilation	Various cells of the gut such as epithelial cells, immune cells, and enteroendocrine cells express cortisol receptors and can be directly affected by HPA dysregulation and increased levels of cortisol.
Defense and Repair	Cytokines can trigger HPA axis activation and development of glucocorticoid resistance in prolonged disease.
Energy	Mitochondrial dysfunction can contribute to symptoms of adrenal insufficiency and cortisol insufficiency.
Biotransformation	Maldigestion and malabsorption are associated with impaired digestion and altered nutrient absorption, affecting the bioavailability of vitamins necessary for optimal adrenal function such as B-complex vitamins, vitamin C, and magnesium.

Modifiable Lifestyle Factors and Mechanisms of Adrenal Dysregulation

Sleep	Sleep deprivation is associated with dysregulated cortisol secretion patterns the next day.
Exercise and Movement	Some studies have found that light to moderate physical activity may beneficially modulate cortisol and sleep quality.
Nutrition	Chronic overconsumption of high-sugar diets can dysregulate HPA axis function. Diets high in added sugars are linked to metabolic disturbances that affect cortisol regulation over time.
Stress	Chronic stress is associated with dysregulation of the HPA axis and increased levels of adrenal hormones and can contribute to type 2 diabetes.
Relationships	Loneliness, also termed perceived social isolation (PSI), is a form of chronic stress that has been linked to disruptions in the HPA axis that lead to changes in cortisol production and regulation.

The Vicious Cycle:

Stress, Adrenal Dysregulation, and Lifestyle Behavior Change

Stress is a mediator of adrenal dysregulation.

Adrenal dysregulation is mitigated by lifestyle behavior changes.

Individuals with high levels of stress find it more challenging to initiate and sustain positive lifestyle behavior changes.

Clinical Focus:

- Correlations of adrenal dysregulation with sex hormone and thyroid hormone imbalances.



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

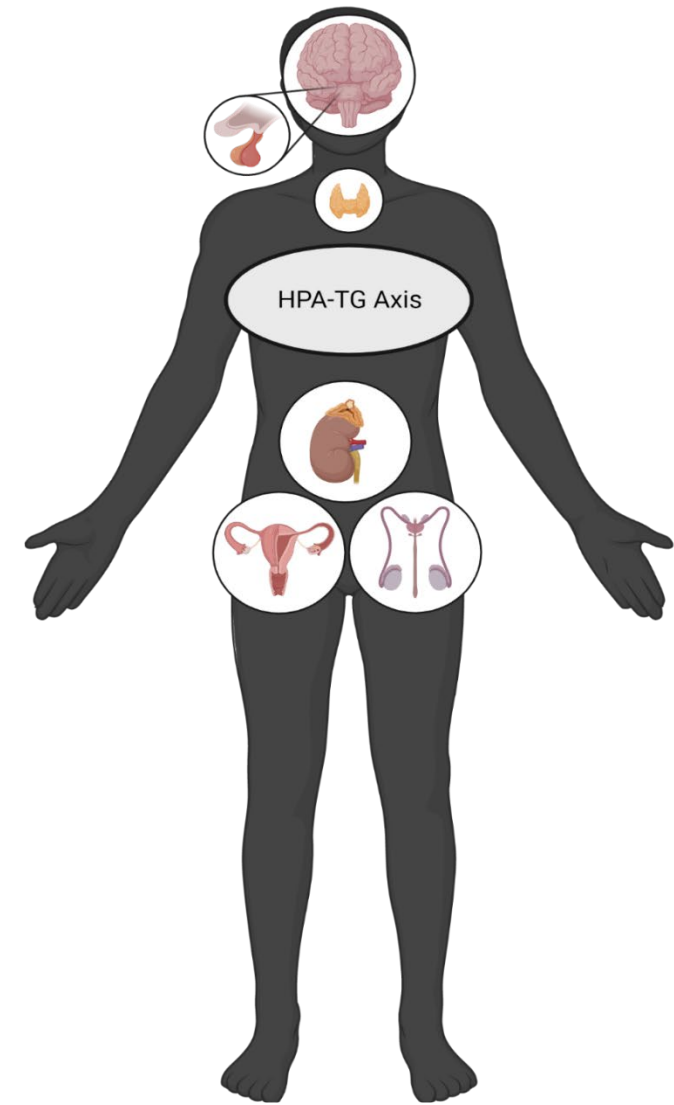
Communication

Mitigation & Clearance

Transport

Systemic Effects
of Adrenal
Dysregulation and
the Chronic
Stress Response

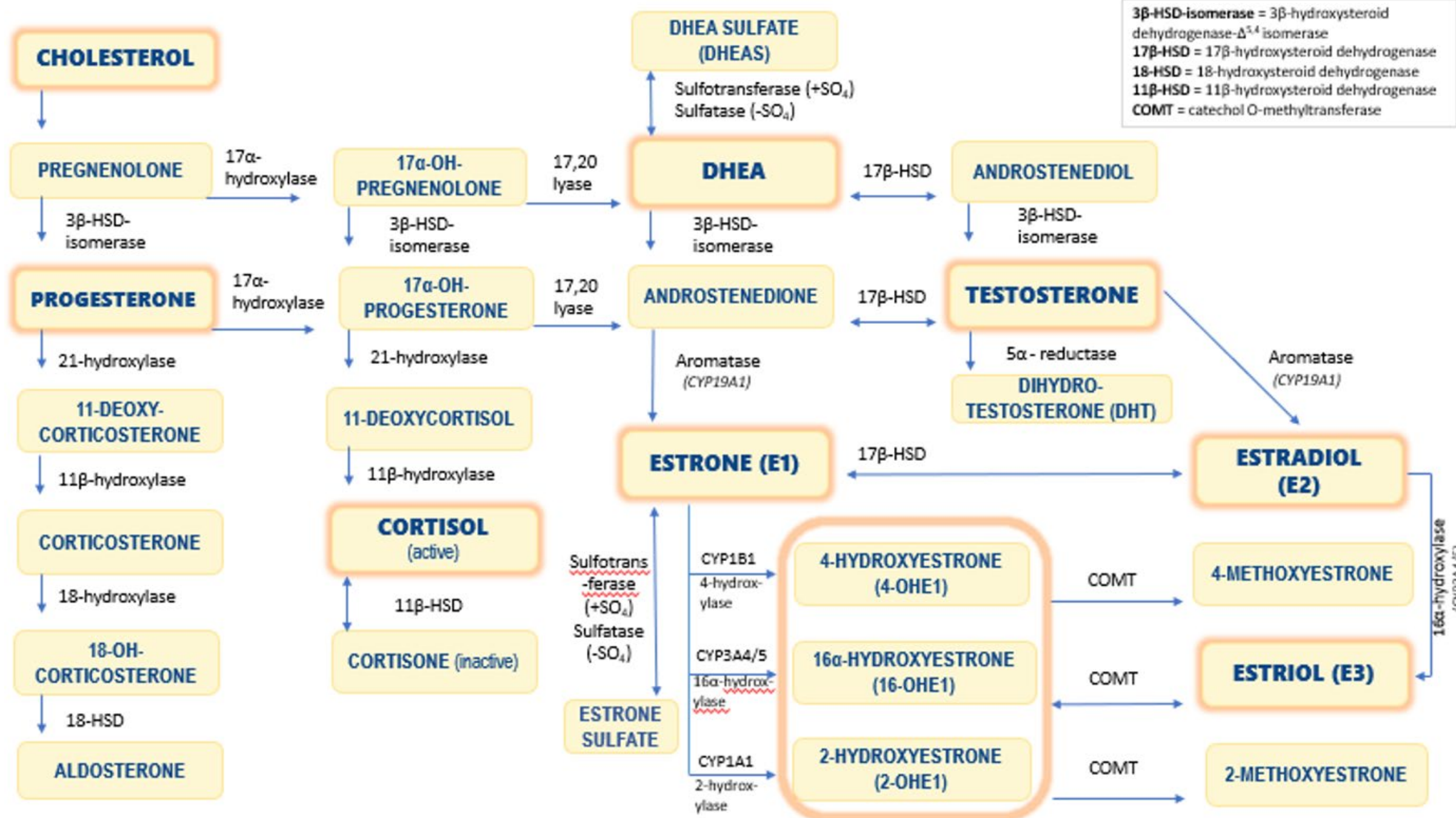
Stress is a central player
in functional alterations
of the hypothalamic-
pituitary-adrenal-thyroid-
gonadal (HPATG) axis.



Created with BioRender.com by The Institute for Functional Medicine

Adrenal Cortisol and Sex Hormone Regulation: Correlations With Stress

- **Stress** is arguably the most common mediator of increased cortisol production.
- The counter-regulatory mechanism affects sex hormones.
- **Stress (and inflammation)** inhibits sex steroid hormone production directly and indirectly (i.e., inhibition of 17,20-lyase).



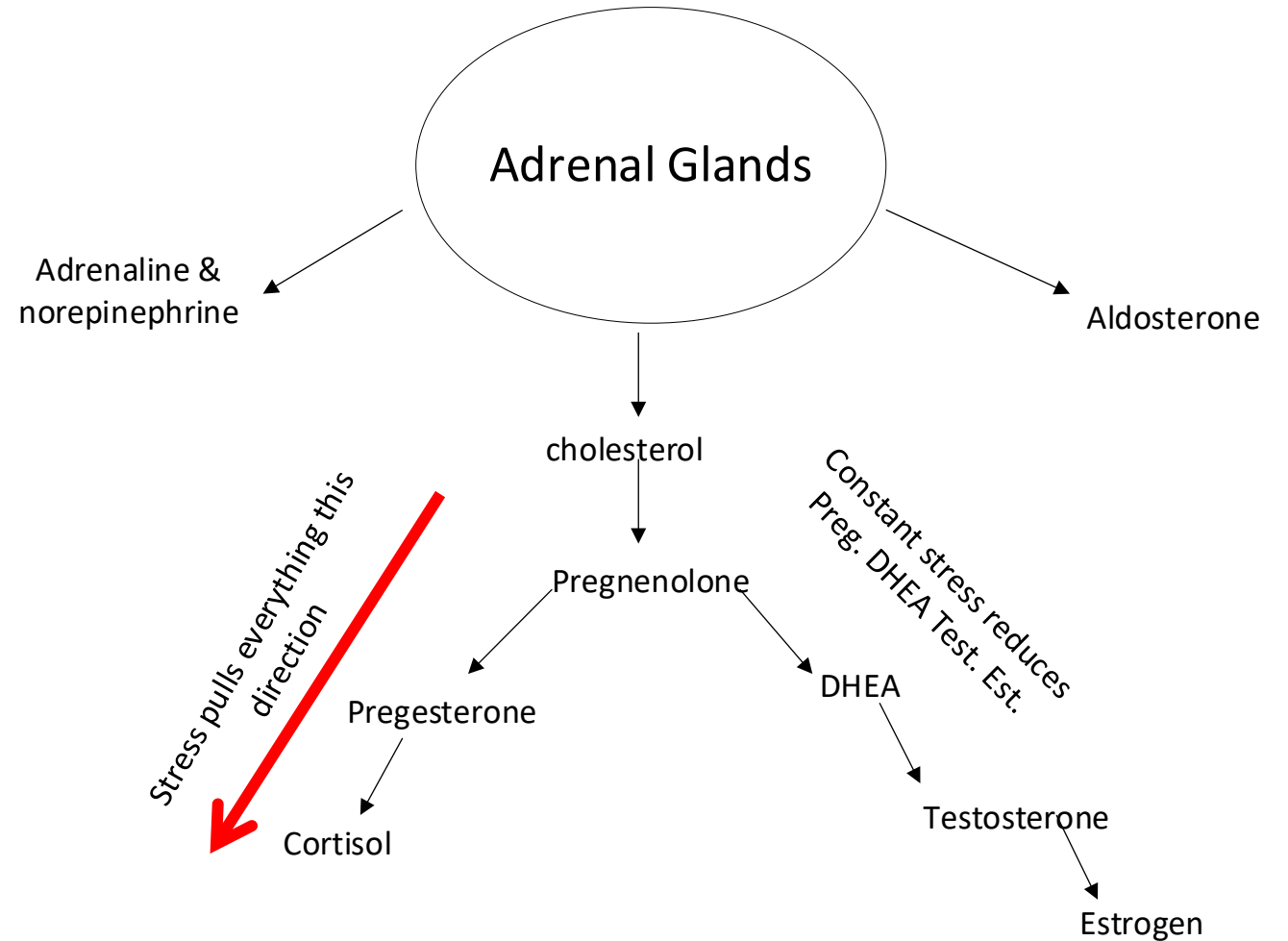
Long Description: Steroidogenic Pathways

- The complex steroid hormone biosynthesis pathways starting from cholesterol. Cholesterol is converted into steroid hormones, including corticosteroids, androgens, and estrogens, through multiple intermediate compounds.
- Cholesterol is converted into **pregnenolone**, which branches into two main pathways:
 - **Corticosteroid Pathway:**
 - Pregnenolone, progesterone, 11-deoxycorticosterone, corticosterone, 18- O H -corticosterone, aldosterone.
 - Progesterone converts into 17 α – O H -progesterone, then 11-deoxycortisol, then cortisol and bidirectionally with cortisone.
 - **Sex Hormone Pathway:**
 - Pregnenolone, 17 α – O H -pregnenolone, D H E A, androstenedione, testosterone.
 - Testosterone, dihydrotestosterone (D H T) via 5 α -reductase.
 - Androstenedione and testosterone can be converted to estrone and estradiol respectively by aromatase.
 - Estrone and estradiol interconvert and can also produce estriol
- **Metabolism of estrogens:** Estrone is hydroxylated into three metabolites:
 - 4-hydroxyestrone
 - 16 A - hydroxyestrone
 - 2 - hydroxyestrone
 - These are further methylated by the enzyme COMT into methoxyestrogens
 - Estradiol converts to estriol via 16 α -hydroxylation and to 2-methoxyestrone.
- **Sulfation and desulfation** also occur:
 - D H E A and estrone can be sulfated into D H E A sulfate and estrone sulfate, respectively, and reversed by sulfatases.
- Enzymes involved include hydroxylases, reductases, isomerases, aromatase, and C O M T
- **Overall Meaning:**

This visual depicts how a single molecule, cholesterol, gives rise to critical hormones that regulate stress response, metabolism, fluid balance, and reproductive function through multiple biochemical steps. It emphasizes the interconnectedness of adrenal and gonadal hormone synthesis.

Adrenal Cortisol and Sex Hormone Regulation

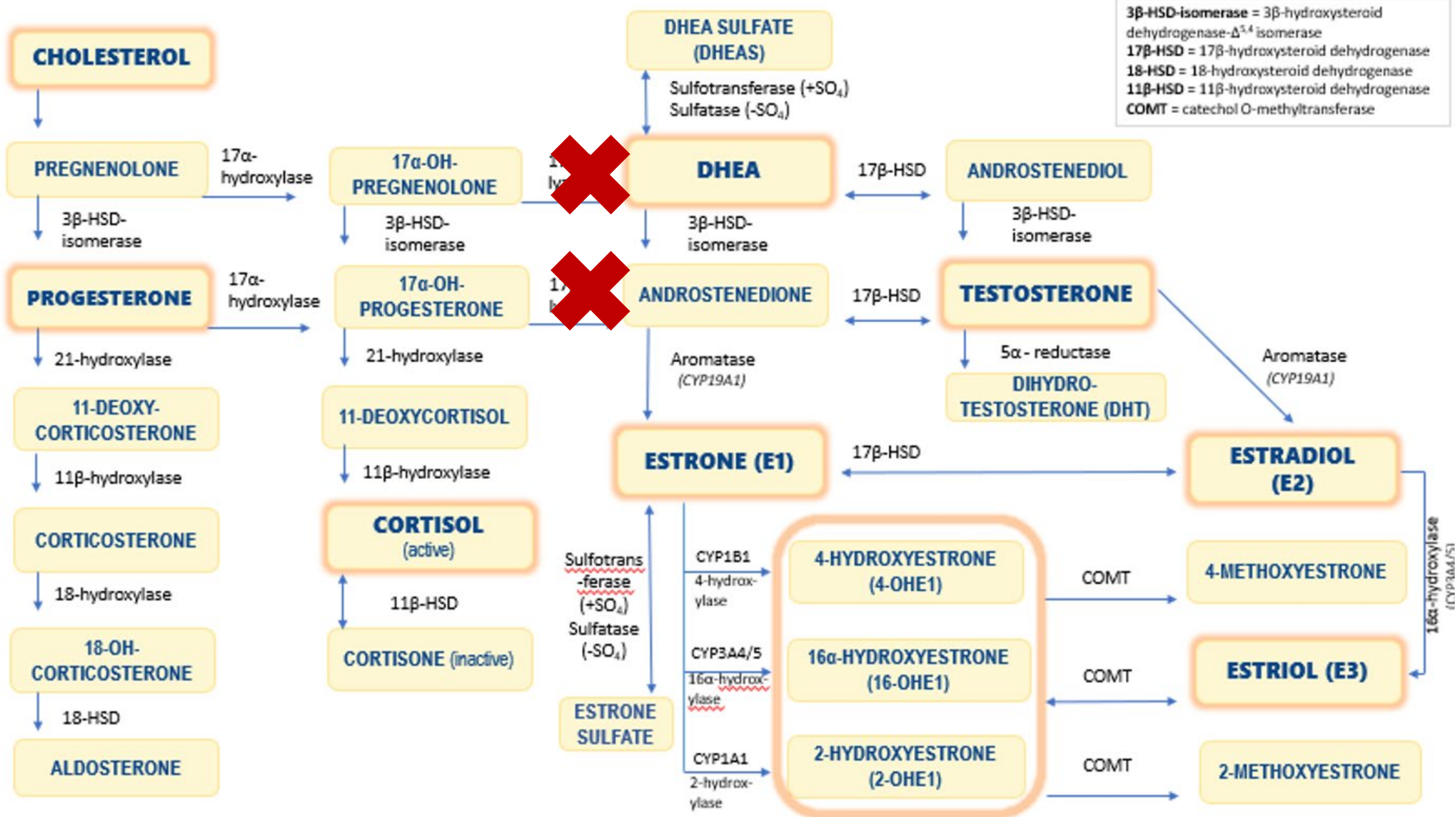
When cortisol production is increased, there is a counter-regulatory decrease in the production and activity of progesterone, estrogens, DHEA, and testosterone.



Long Description:

Adrenal Cortisol and Sex Hormone Regulation

- This diagram illustrates how the adrenal glands produce various hormones and how chronic stress influences hormone pathways.
- The adrenal glands produce three main types of hormones:
 - **Adrenaline and norepinephrine**
 - **Aldosterone**
 - **Cholesterol**, which is used as a precursor for steroid hormone synthesis.
- From **cholesterol**, the hormone production pathway branches:
- **Cholesterol** converts into **pregnenolone**.
 - Pregnenolone can follow two main pathways:
 - Convert into progesterone, which then leads to cortisol.
 - Convert into DHEA, which can further convert into testosterone and then estrogen.
- A **large red arrow** points strongly toward the **progesterone-to-cortisol** pathway with the text: *"Stress pulls everything this direction"*, implying that under chronic stress, the body prioritizes cortisol production.
- Another note reads: *"Constant stress reduces Pregnenolone, DHEA, Testosterone, Estrogen"*, indicating that ongoing stress depletes resources needed for sex hormone production.
- **Key Message:**
Chronic stress causes the body to divert hormone production toward cortisol at the expense of sex hormones like DHEA, testosterone, and estrogen, leading to hormonal imbalances.



Long Description:

Steroidogenic pathway showing inhibition of 17,20 lyase

- This diagram illustrates the steroid hormone biosynthesis pathway and highlights the **inhibition of the 17,20 lyase enzyme**, which is critical for producing sex hormones.
- The pathway begins with **cholesterol**, which is converted to **pregnenolone**, then progresses through several enzymatic steps into three main hormonal branches:
 - **Corticosteroids** (e.g., cortisol and aldosterone)
 - **Androgens** (e.g., DHEA, testosterone)
 - **Estrogens** (e.g., estradiol, estriol)
- Two **red X marks** indicate that the **17,20 lyase enzyme** activity is **blocked** at two key branch points:
 - Conversion of **17 alpha hydroxypregnenolone to DHEA**
 - Conversion of **17 alpha hydroxyprogesterone to androstenedione**
- **As a result of this inhibition:**
 - **DHEA, androstenedione, and testosterone** production is significantly reduced.
 - **Downstream estrogen synthesis** (estrone, estradiol, estriol) is also impaired.
 - The body is pushed toward producing **cortisol** and other corticosteroids instead.
- **Key Message:**

When 17,20 lyase is inhibited, the body's ability to synthesize sex hormones (DHEA, testosterone, estrogens) is impaired, shifting hormone production toward the cortisol and aldosterone pathways. This reflects how enzyme dysfunction or chronic stress can lead to hormonal imbalances.

Physiological Effects of Increased Cortisol Production on Sex Hormone Balance

Short-Term Effects

Reduced progesterone production.

Stimulation of aromatase resulting initially in an estrogen-dominant state.

May increase risk for estrogen-dominant conditions:

- Breast cancer
- Fibroids
- Endometriosis

Long-Term Effects

Decreased formation of androgens and estrogens (inhibition of DHEA pathway to form androgens and estrogens).

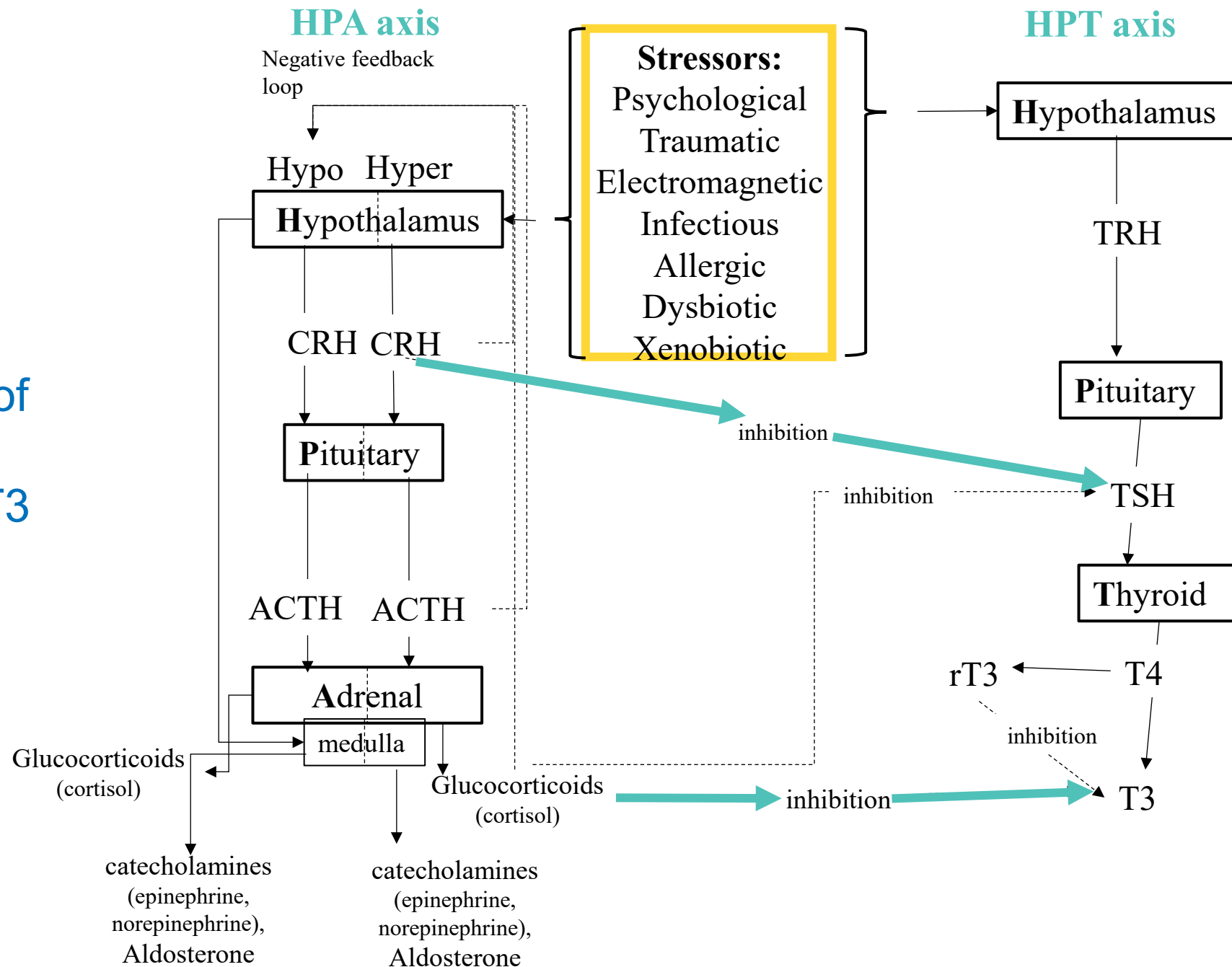
Explains associations between stress & hot flashes, stress & decreased libido.

References

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Mechanisms of Adrenal Dysregulation on Thyroid Function: Impaired Production of TSH and Impaired Conversion of T4 to T3

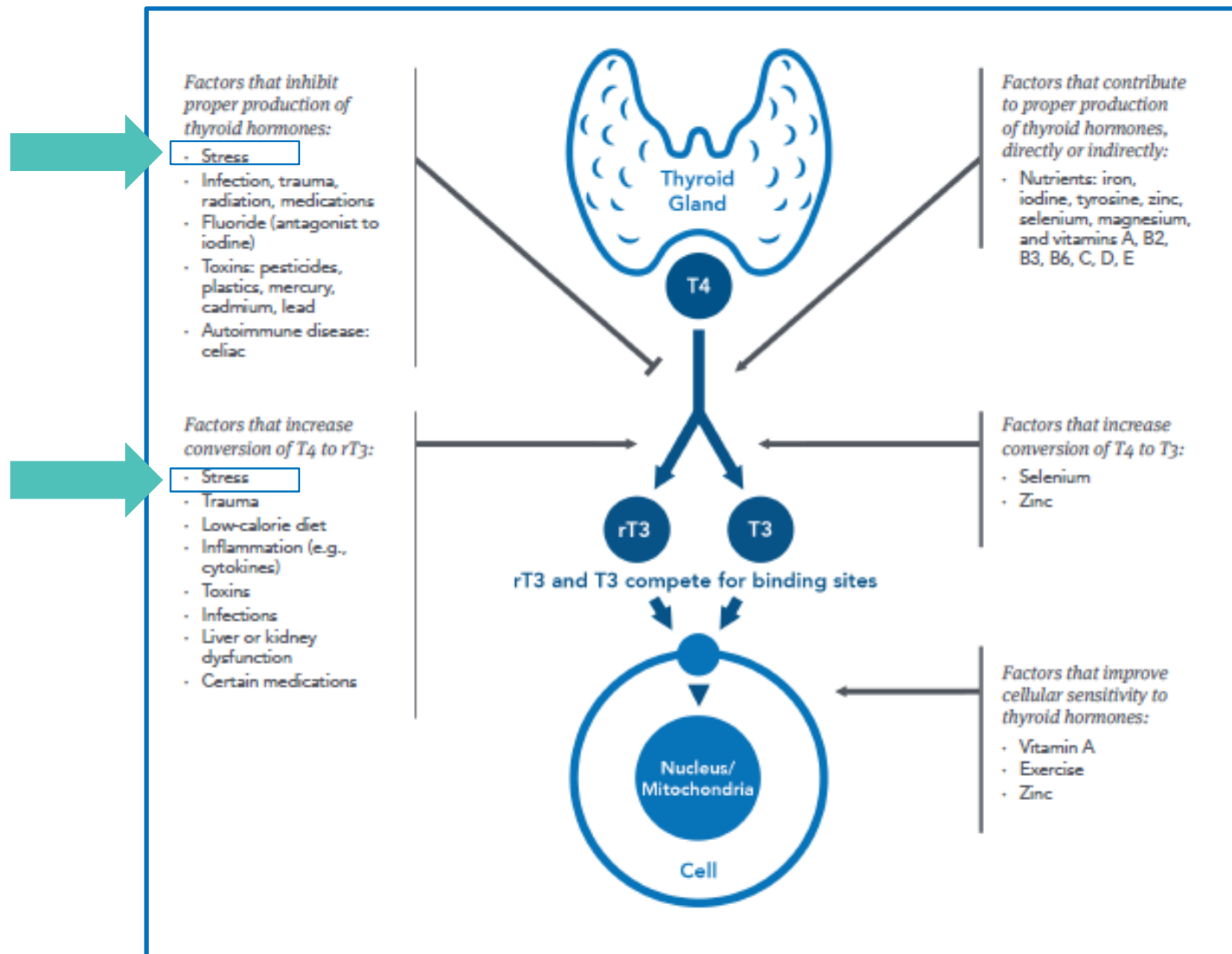


Long Description:

Mechanisms of Adrenal Dysregulation on Thyroid Function:

Impaired Production of T S H and Impaired Conversion of T 4 to T 3

- The diagram illustrates how **adrenal dysregulation impairs thyroid function**, highlighting the interaction between the **H P A axis** (hypothalamic-pituitary-adrenal) and **H P T axis** (hypothalamic-pituitary-thyroid).
- **Left Side: H P A Axis**
 - **Stressors** - including psychological, traumatic, electromagnetic, infectious, allergic, dysbiotic, and xenobiotic factors - activate the **HPA axis**.
 - The **hypothalamus** releases **C R H** (corticotropin-releasing hormone), which signals the **pituitary** to release **A C T H** (adrenocorticotrophic hormone).
 - A C T H stimulates the **adrenal glands**, leading to the release of:
 - **Glucocorticoids (cortisol)**
 - **Catecholamines (epinephrine and norepinephrine)**
 - **Aldosterone**
- **Right Side: H P T Axis**
 - The **hypothalamus** releases **T R H** (thyrotropin-releasing hormone), which stimulates the **pituitary** to release **T S H** (thyroid-stimulating hormone).
 - TSH prompts the **thyroid** to produce **T4 (thyroxine)**, which must be converted into the active form **T3 (triiodothyronine)**.
 - Under stress, **cortisol** inhibits:
 - T S H production in the pituitary
 - The conversion of T4 to T3 in the body
 - Instead, T4 may be converted to **r T3 (reverse T3)**, an inactive form.
- **Key Message - Adrenal dysregulation caused by chronic stress suppresses thyroid function in two ways:**
 - By lowering TSH production.
 - By blocking the conversion of T4 into its active form T3.
 - This leads to reduced thyroid hormone activity even if T4 levels appear normal.



Stress & Autoimmune Thyroid Disease

- **Autoimmune thyroid conditions are the most influenced by stress** (Hashimoto's and Grave's disease).
- Research has found a link between stressful life events and the onset of Graves' disease.

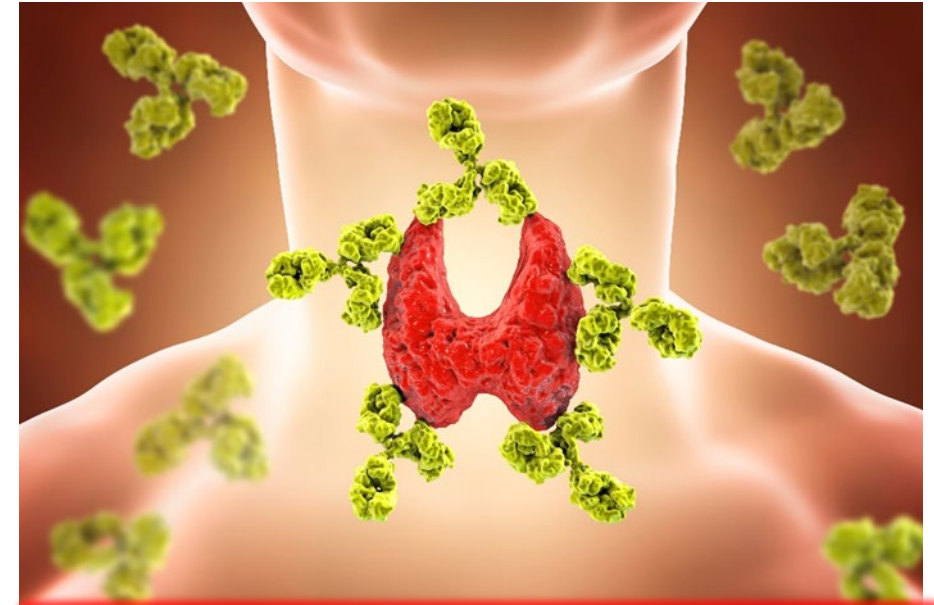


Stress

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Stress & Autoimmune Thyroid Disease

A correlation has also been found between self-reported stress and disease progression, suggesting that **stress management may be effective in improving the prognosis of Graves' hyperthyroidism.**



Autoimmune Thyroiditis

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STRESS & AUTOIMMUNE THYROID DISEASE

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Communication and Hormone Balance: Part 3

Addressing Adrenal Dysregulation and Chronic Stress

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus:

- Foundational therapeutic interventions for adrenal dysregulation.



Adrenal Dysregulation: A Functional Medicine Approach

- Align the physiological priorities (from patterns identified on the matrix) with the patient's preferences and goals.
- Informed by these two parameters, initiate interventions that are feasible for the patient.

“What matters to the patient?”



Helpful Referrals for Adrenal Dysregulation

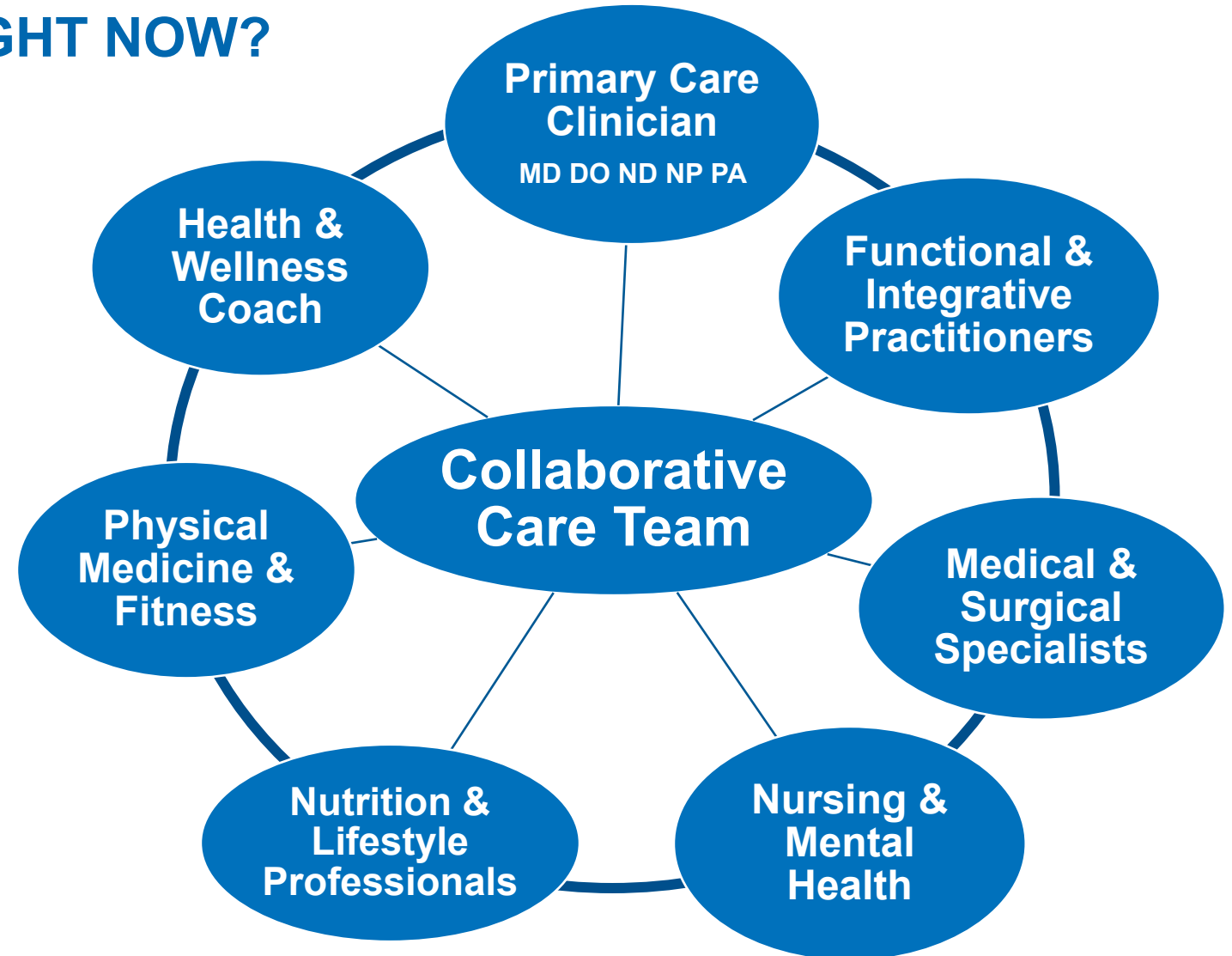
- **Social Services**
 - **Coaches**
- **Nutrition Professionals**
 - **Counseling**
- **Mental Health Professionals**
- **Integrative Care Providers**
 - **Others**

Getting Healthy Can Take a Team Approach



The Collaborative Care Team

WHO IS ON YOUR TEAM RIGHT NOW?



Foundational Therapeutic Approach: Chronic Stress & Adrenal Dysfunction/ Communication & Hormone Balance



LIFESTYLE FIRST



FOOD FIRST



GUT FIRST

Modifiable Lifestyle Factors:

Essential Interventions to Restore Adrenal Function



Overall Lifestyle Changes



Key Questions to Rapidly Elicit “Lifestyle Mediators”

- Which of your regular habits, behaviors, and pastimes:
 - ...have a negative effect on your health?
 - ...have a positive effect on your health?
- How would you answer these questions?
 - My health would be better if I spent less time _____.
 - My health would be better if I spent more time _____.

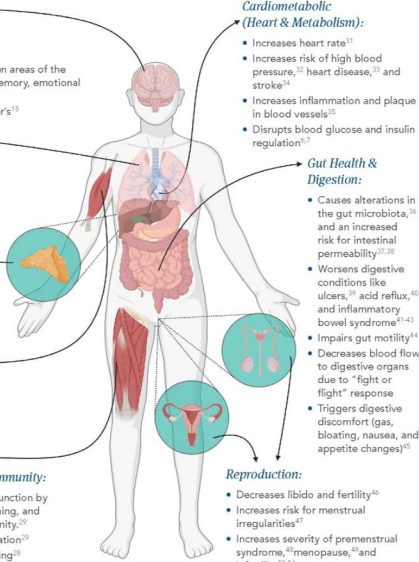
Educate Your Patient

In Your Toolkit



Effects of Chronic Stress

Chronic stress refers to prolonged periods of mental stress, which can have lasting effects on the entire body. People who report higher levels of chronic stress are more likely to develop conditions including heart disease,¹⁻³ depression,^{4,5} diabetes,^{6,7} dementia,^{8,9} cancer,¹⁰⁻¹² and digestive symptoms.^{13,14} The following diagram shows potential harmful changes in the body as a result of unaddressed chronic stress.^{15,17}



Brain & Nervous System:

- Activates "fight or flight" response¹⁸
- Reduces size and communication between areas of the brain responsible for decision making, memory, emotional regulation, and learning^{13,14}
- Increases risk for dementia and Alzheimer's¹⁵

Respiratory System:

- Causes changes in breathing patterns that shift to rapid, shallow breathing as a part of the fight-or-flight response^{16,17}
- Worsens asthma and chronic obstructive lung disease^{18,19}

Hormone Regulation:

- Causes thyroid hormone fluctuations²⁰
- Releases more cortisol (a stress hormone)^{21,22} which increases the risk of other health conditions^{23,24}

Musculoskeletal:

- Contributes to muscle tension
- Increases risk of headaches and migraines²⁵
- Worsens joint pain and fibromyalgia^{26,27}

Inflammation & Immunity:

- Impacts immune function by activating, weakening, and suppressing immunity²⁸
- Increases inflammation²⁹
- Slows wound healing³⁰

Cardiometabolic (Heart & Metabolism):

- Increases heart rate³¹
- Increases risk of high blood pressure,³² heart disease,³³ and stroke³⁴
- Increases inflammation and plaque in blood vessels³⁵
- Disrupts blood glucose and insulin regulation³⁶

Gut Health & Digestion:

- Causes alterations in the gut microbiota,³⁸ and an increased risk for intestinal permeability^{37,38}
- Worsens digestive conditions like ulcers,³⁹ acid reflux,⁴⁰ and inflammatory bowel syndrome⁴¹⁻⁴³
- Impairs gut motility⁴⁴
- Decreases blood flow to digestive organs due to "fight or flight" response
- Triggers digestive discomfort (gas, bloating, nausea, and appetite changes)⁴⁵

Reproduction:

- Decreases libido and fertility⁴⁶
- Increases risk for menstrual irregularities⁴⁷
- Increases severity of premenstrual syndrome,⁴⁸ menopause,⁴⁹ and infertility^{50,51}

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Common Chronic Stressors

- Information overload from social media and news
- Relationship conflicts
- Expectations to multitask at work and home
- Social isolation
- Finances
- Managing chronic illness
- Traumatic events (e.g., death of a loved one, COVID-19)

Tips for Reducing Chronic Stress

Day-to-day stress may be unavoidable, but it does not have to turn into chronic unaddressed stress that negatively impacts your health. There are many ways to manage stress – from daily activities like deep breathing to longer-term strategies like counseling. Finding a variety of techniques that work for you will help you build resilience and support your overall wellbeing for years to come. Here are more ideas to help you manage stress:

- **Deep Breathing:** Try breathing exercises to encourage deep breathing, which activates your relaxation response⁵²
- **Adequate Rest:** Prioritize getting adequate sleep. For adults, this means at least 7 to 9 hours of quality sleep each night^{53,54}
- **Exercise:** Try exercise that combines physical movement with mental relaxation.^{55,56} This includes aerobic exercises such as walking,⁵⁷ swimming,⁵⁸ cycling,⁵⁹ dancing,⁶⁰ tai chi,⁶¹ Qi gong,⁶² yoga,^{63,64} and strength training⁶⁵
- **Therapy:** Talk to a counselor or therapist. Ask your functional medicine practitioner for a referral
- **Journaling:** Start a gratitude journal. Writing down things that you are grateful for can decrease stress and improve mood^{66,67}
- **Meditate:** Meditating can help relieve stress.⁶⁸ There are many ways to meditate, so find what works for you.
- **Take a Mindfulness-Based Stress Reduction (MBSR) course:** Search online for virtual or in person courses near you^{64,69}
- **Spend Time Around Nature:** Spending time in green spaces and natural environments can decrease stress.⁷⁰ Here are some ideas for increasing your exposure to nature:
 - Get outside for a walk or go for a hike at a park. Start a garden outside^{71,72}
 - If you are unable to get outside, use images of nature, like landscapes or greenery, as your computer background or as art in your home or workspace^{73,74}
 - Create a green space in your home. Consider adding a plant to your space indoors^{74,75}

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- **Digital detox:** Do a digital detox. Limit screen daily screen time or try to avoid electronic devices for 24 hours, or even a whole weekend^{76,77}
- **Reframe your self-talk:** Changing the way we think about things that happen can help decrease our stress response to them. For example, instead of thinking "They rejected my job application," think, "I wasn't rejected. I'm being redirected toward something better"^{78,79}
- **Create and reassess your "to-do" list:** Prioritize your top 3 tasks and shift your expectations. Ask for help or hand-off tasks others can take on
- **Optimize your nutrition:** Nutrition can help support a healthy stress response. Prioritize healthy protein and fats. Eat a variety of colorful vegetables and fruits⁸⁰⁻⁸²
- **Socialize with supportive friends or family:** Having people who listen and care can help decrease stressful situations
- **Consider music therapy:** Listening to different types of music creates different effects in the body. With music therapy, music is tailored to decrease stress by creating calmness and relaxation. Search for a practitioner near you who specializes in this⁸³

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Therapeutic Modalities for Patients



Practicing Gratitude with the IFM Gratitude Journal

"When I started counting my blessings, my whole life turned around."
Willie Nelson

In recent years, the act of recognizing and expressing gratitude has been linked to many improved physical, psychological, and social health benefits for both healthcare providers and patients. These benefits include decreased stress levels, lowered blood pressure, better sleep quality, stronger immune systems, and increased feelings of joy, happiness, forgiveness, and compassion. Robert A. Emmons, Ph.D., a leading scientific expert on gratitude, sees the practice of gratitude as having two components: affirming the goodness in our lives, and exploring where that goodness comes from.

Here are some tips for practicing and exploring gratitude using a gratitude journal:

- Spend at least 15 minutes writing about the things for which you are grateful. You may find it easier to write at night, so you can include events or observations from the day.
- Set an alarm on your watch, calendar, or smart phone to help remind you to write in your journal once or twice per week. Try to integrate journaling into your routine, but also write down joyful experiences as they happen so that you don't forget them.
- Begin your journey by writing about the basic material things about which you are grateful—such as having a home, a warm bed, food, etc. Then, try to shift the focus to the people, places, experiences, and situations that bring you joy and happiness.
- Avoid repeating the same entries. This will make journaling a challenge as times pass, but challenging yourself in this way will allow your awareness and sense of gratitude to flourish.
- Instead of tallying up all of the good people and things in your life, reflect deeply on how these people and things bring you joy, and what your life would be like without them.
- Try to find something to appreciate in the people and things you don't like. Focusing only on the goodness will help shift your perceptions and bring you more happiness.

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Meditation: How to Get Started

Meditation can support your physical, mental, and emotional well-being. There are many types of meditation, but it can be as simple as sitting and focusing on your breathing.

STUDIES SUGGEST MEDITATION MAY HELP:¹⁻¹⁷

- Addictions
- Anxiety
- Attention/focus
- Autoimmune diseases
- Brain function
- Cardiovascular health
- Depression
- Diabetes
- Fatigue
- Glaucoma
- Immune system function
- Irritable bowel syndrome
- Pain
- Post-traumatic stress disorder (PTSD)
- Sleep
- Stress

What Is Meditation?

Meditation is an umbrella term for mind-body practices that help calm your mind, focus your attention, and heighten internal awareness.¹⁸⁻²⁰ Several forms of meditation have been practiced in various cultures for centuries.

According to national surveys, around 15% of adults meditate in countries including the United States, Britain, and Germany.¹⁵⁻¹⁷ In India, where certain types of meditation originated, an average of 48% of adults meditate.¹¹

Though there are many different kinds of meditation, they tend to share common elements, including:¹⁸

- **Minimized distractions.** You don't need complete silence. However, you should be free of distractions trying to interact with you, such as other people or your phone.
- **Focused attention.** You might use a word or phrase, a physical object, a visualized object, sounds, or your breathing to help focus your attention.
- **Open attitude.** You aim to let distracting thoughts, emotions, or physical symptoms come and go without analyzing or judging them.

One of the most commonly practiced types of meditation is mindfulness meditation.²¹ It is designed to help you focus on the present moment.¹⁸ This can help bring about physical and mental relaxation, but that is just one of the many ways meditation may help you.²¹

Why Meditate?

When you quiet your mind through meditation, it could have far-reaching benefits on your health. It may do this in several ways, including:

- **Reducing cortisol.** This stress hormone stays at high levels when stressful events frequently occur even after the stressful event ends.²²⁻²⁵ That increases your risk of dysfunction and disease.

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Strategies for Transforming Stress

You can't avoid stress, but you can manage it. Many simple relaxation strategies can help you transform stress. When you practice them every day, they will become healthy habits. Learning to adapt well in the face of challenges can bolster your energy reserves, support your health, and improve your quality of life.

Why Stress Relief Matters

Stress is one of the top health concerns of the 21st century—and for a good reason. The majority of visits to primary care doctors are due, in part, to the effects of chronic stress. For example, prolonged stress has been linked to health issues like heart disease, diabetes, and chronic pain. Ongoing stress can also harm your mood, emotions, and energy level.

Studies suggest that practicing relaxation methods can help you cope in a wide variety of situations and may improve health-related symptoms. Some of the symptoms and conditions that stress-relief practices can help with include:

• Anxiety	• Heart disease	• Parkinson's disease
• Cancer (during and after treatment)	• High blood pressure (hypertension)	• Pre-surgery stress
• Caregiver stress	• Insomnia	• Quitting smoking
• Childbirth	• Job-related stress	• School-related stress
• Depression	• Multiple sclerosis	• Temporomandibular joint dysfunction (TMJ) or jaw pain
• Diabetes	• Pain control	
• Headaches, migraines	• Parenting stress	

Ways to Manage Stress

Many proven stress-relief approaches are listed here. Choose the methods that appeal to you and work them into your daily routine. For example, do 10 minutes of yoga every day during your lunch break. Or, listen to relaxing music for 15 minutes every night before bed.

- Go for a walk or do other physical activities you enjoy
- Journal about things you are grateful for
- Use a free gratitude app online

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Therapeutic Modalities for Patients



The Power of Movement: Living an Active Lifestyle

"Lack of activity destroys the good condition of every human being, while movement and methodical physical exercise save it and preserve it." —Plato

Being consistently active helps you to live longer, have a better quality of life, improve your mental health, and improve your self-image. Adults should aim for either 150-300 minutes of moderate-intensity activity per week, 75 minutes of high-intensity activity, or a combination of both.

You can make big improvements to your health and energy levels by incorporating exercise and physical activity throughout your week. Some people feel daunted by the thought of adding physical activity or don't know where to begin. Here are a few tips to support your journey:

Emphasize fun. What is something you love? Whether that's music, birds, friends, trampolines, or books, you can shape your activity plan around the things you love. Walk to and from a spot where you can listen to the birds every day; explore local libraries from top to bottom; take the stairs when you visit friends; take a dance class that incorporates music you love. Make the things you love part of your activity plan.

Create a plan Create a plan and carve out time for physical activity and exercise throughout the week. If you block time on your schedule for activity, it is much more likely to happen. Spread your exercise minutes throughout the week. Two 20-minute high-intensity interval training sessions, a one hour family bike ride, and a 20 minute jog is a good example of how you can meet the amount of activity recommended.

Involve others. Chances are, your friends, family, and co-workers want to be more active too. Set active living goals together and aim for gradual advancement and increased variety in your routine. You could walk an extra two miles a week—or three more flights of stairs. Try to keep moving by walking instead of sitting at the coffeeshop, or walk around the block while catching up with a friend. Many cities have active groups like running or hiking clubs which can be great in your home town or when traveling to keep you moving.

Add an accountability partner. Share your goals with a person you trust and ask them to help keep you accountable by checking in. It can be helpful to partner with a friend or family member who has similar interests to engage in activity together and keep each other accountable. Online groups may also



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Suggestions for Better Sleep

Achieving deep, refreshing sleep isn't always as simple as cozying up in bed and closing your eyes – it often depends on your daily lifestyle choices and habits. From how you unwind before bed to what you eat, small adjustments can make a big difference in your sleep. This guide walks you through suggestions that can help you achieve better sleep, so you can wake up feeling well-rested, energetic, and ready to start your day.

Step 1: Tune Up Your Sleep Hygiene

Sleep hygiene is a set of healthy habits and environmental conditions that can help you sleep soundly. If you are struggling with your sleep, tuning up your sleep hygiene is the place to start. Try the following sleep hygiene tune-ups for better sleep:

- **Go to bed and wake up at the same time daily.** A consistent waking time and bedtime each day supports your circadian rhythm, your body's internal 24-hour clock that regulates sleep and other body functions. A healthy circadian rhythm helps you stay alert and energetic during the day and eases you to sleep at night.¹ Ideally, your bedtime and wake time should be set to allow for 7-9 hours of sleep each night.²
- **Sleep in a dark bedroom.** Darkness promotes sleep by supporting a healthy circadian rhythm. Even dim artificial light exposure at night from sources like streetlights can interfere with melatonin, a hormone that regulates sleep.^{3,4} If you have lights outside your bedroom at night or work the night shift and sleep during the day, hang blackout curtains on your bedroom windows or wear an eye mask. Remove electronics that have bright lights from your bedroom. For example, replace a glowing digital alarm clock with a non-digital alarm clock.
- **Avoid spending time on electronic devices two hours before bed.**^{5,6} The bright light from electronic devices such as smartphones and tablets can increase alertness and reduce sleepiness, making it harder to drift off to sleep.^{5,7} Pick an activity that doesn't involve electronic devices before bed. For example, read a hardcover or paperback book.⁸⁻¹⁰ If you must use electronic devices like eReaders close to bedtime, put the device on "nighttime mode." Wearing blue light-blocking glasses while using electronic devices at night may reduce the negative impact of these devices on sleep.¹¹

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Modifiable Lifestyle Factors

Physical Activity: Mitigation of Adrenal Dysregulation



Some studies indicate that light to moderate physical activity may be beneficial for lowering cortisol levels and improving sleep quality.



Modalities to Address Stress: Relaxation and Mindfulness

- **CBT Techniques**
 - Relaxation training
 - Progressive muscle relaxation
- **Primary Care Interventions**
 - Teach relaxation techniques: breathing, meditation, mindfulness
 - Promote sense of self-awareness w/compassion
 - Promote connectedness with family and friends
 - Promote movement/exercise to tolerance
- **Biofeedback-Based Intervention**
- **Dial UP the self-care one step at a time!**



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Breathe

SLOW
DEEP
INHALE



LONGER
FULL
EXHALE

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Modifiable Lifestyle Factors

Mental Health: Mitigation of Adrenal Dysregulation



- Stress management interventions outperformed control conditions in a meta-analysis study on cortisol.
- Mindfulness, meditation, and relaxation interventions were found to reduce cortisol levels.



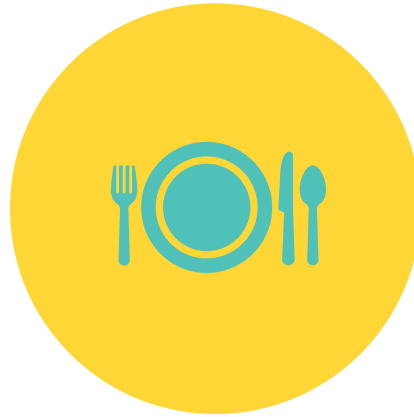
Find Joy and Connection!!



Foundational Therapeutic Approach: Chronic Stress & Adrenal Dysfunction/ Communication & Hormone Balance



LIFESTYLE FIRST



FOOD FIRST



GUT FIRST

Key Questions to Rapidly Elicit “Dietary Mediators”

- How would you answer these questions?
 - My health would be better if I drank less _____.
 - My health would be better if I drank more _____.
- How would you answer these questions?
 - My health would be better if I ate less _____.
 - My health would be better if I ate more _____.

Glycemic Index

- Maintaining a healthy blood sugar level is important for your overall health. Poor blood sugar control can contribute to or worsen chronic metabolic conditions such as diabetes, prediabetes or insulin resistance, hypertension, and heart disease.
- Highly processed diets, which are high in added sugars and refined carbs, can contribute to chronically high insulin levels (hyperinsulemia).
- **Evidence suggests that chronically high insulin levels are associated with HPA axis dysfunction and increased cortisol production.**



Glycemic Index and Glycemic Load

Maintaining healthy blood sugar levels is very important for your overall health. Diabetes, prediabetes or insulin resistance, hypertension, heart disease, and many other chronic conditions are caused or worsened by poor blood sugar control. Not all foods have the same impact on blood sugar and insulin. The Glycemic Index (GI) and Glycemic Load (GL) are two concepts that can help you select foods for optimal blood sugar balance.

The **Glycemic Index (GI)** is a way to measure the impact of a specific food on blood sugar levels. The GI ranks carbohydrate-containing foods on a scale of 0 to 100 based on how quickly the foods raise blood sugar levels. The GI is a score of a food's impact on blood sugar measured in laboratory studies when a specific portion size was eaten. The GI score of a food can vary depending on the food's size, ripeness, storing method, and preparation method.

The **Glycemic Load (GL)** is calculated by multiplying a food's GI (as a percentage) by the number of grams of carbohydrates in a serving size. The result is a score of how much that serving of food is likely to increase blood sugar levels. GL takes into account the GI score and the portion size that is eaten.

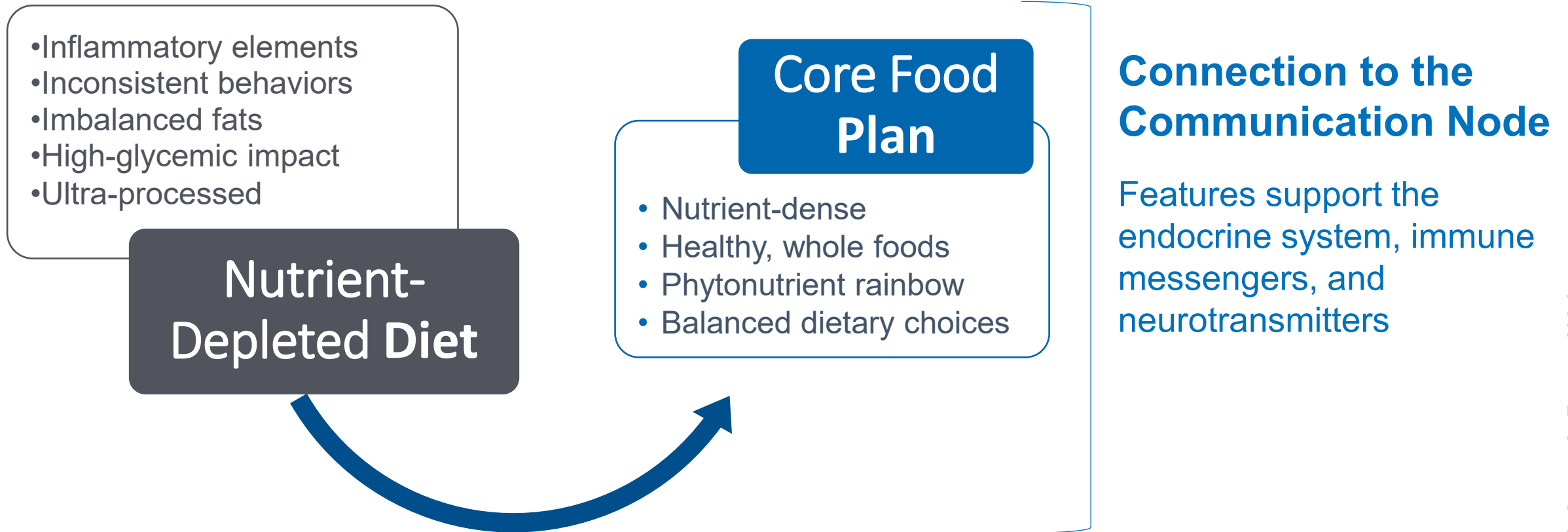
Generally speaking, eating appropriate portions of foods lower in both GI and GL helps to stabilize blood sugar throughout the day. The ranges for low, medium, and high GI and GL are shown in the table below.

	Low	Medium	High
Glycemic Index	55 or less	59–69	70 or higher
Glycemic Load	10 or less	11–19	20 or higher

Tips for Blood Sugar Balance:

- While low GI foods are excellent choices, most people also eat some medium GI foods. One important tip to remember is that **medium and high GI foods should be eaten with protein or fat**, in order to soften the effect of these foods on your blood sugar level. High GI foods are not recommended on a daily basis, as eating these frequently can result in blood sugar imbalances.

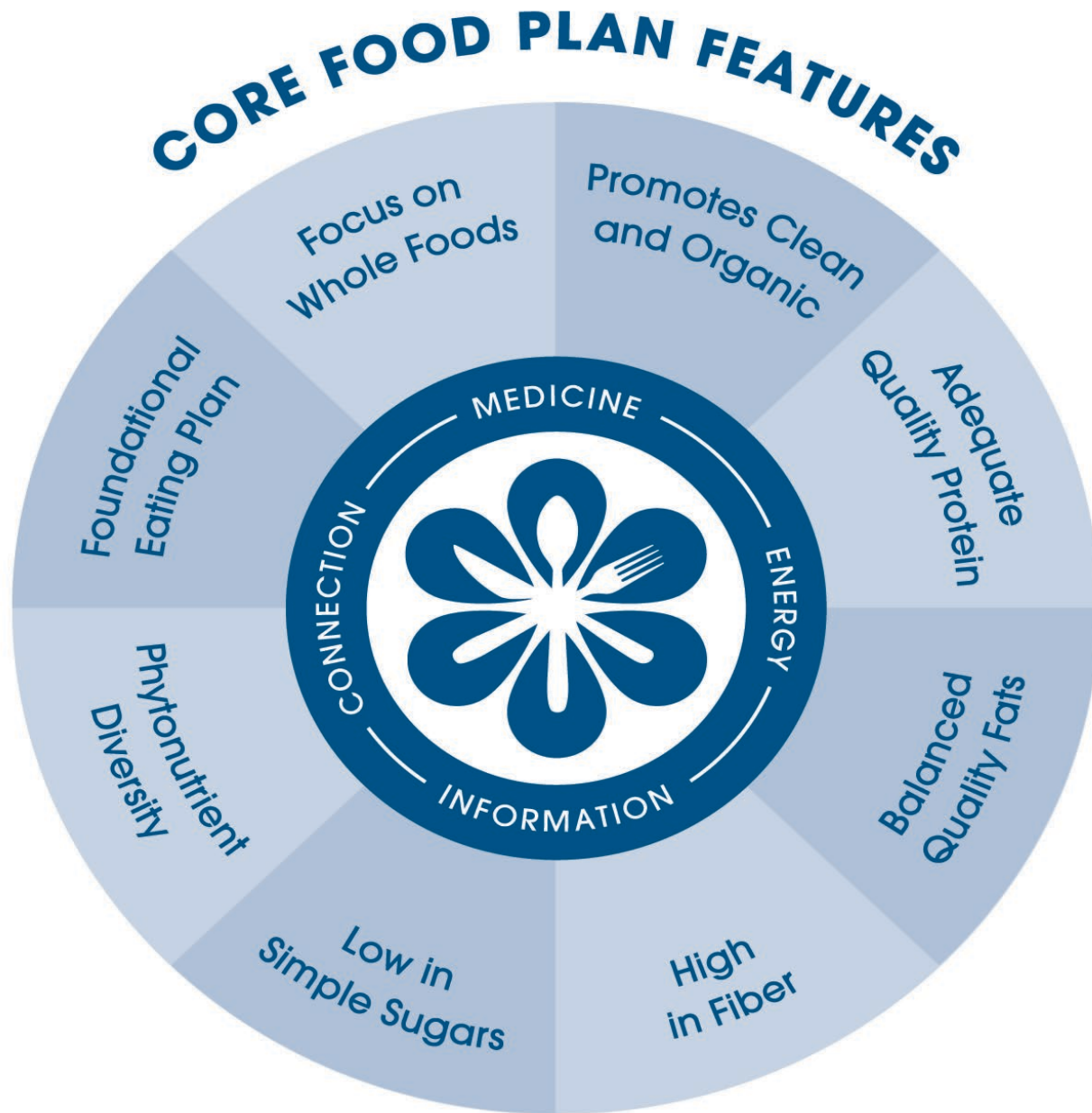
Recommended “Core” First Step for Most Patients: A More Nutrient-Dense Diet



Long Description -

Recommended “Core” First Step for Most Patients: A More Nutrient-Dense Diet

- The diagram illustrates a recommended foundational intervention: shifting from a **nutrient-depleted diet** to using the **Core Food Plan** as a first step for most patients.
- On the **left**, a **Nutrient-Depleted Diet** is described as having:
 - Inflammatory elements
 - Inconsistent behaviors
 - Imbalanced fats
 - High-glycemic impact
 - Ultra-processed foods
- An arrow curves from this depleted diet toward the **Core Food Plan**, indicating a positive transition.
- The **Core Food Plan** emphasizes:
 - Nutrient-dense foods
 - Healthy, whole foods
 - A variety of colorful plant foods (“phytonutrient rainbow”)
 - Balanced dietary choices
- On the **right**, this plan is connected to the “**Communication Node**”, explaining that:
 - Core food plan features support the **endocrine system**, **immune messengers**, and **neurotransmitters**.



Phytonutrient *Spectrum*



Comprehensive Guide

Food-Based Foundational Nutrients to Maintain Adrenal Health

- **B Vitamins** (cofactors in hormone production):
 - **B5 – Pantothenic Acid:** Liver, egg yolks, chicken, salmon, dairy
 - **B6 – Pyridoxine:** Chicken breast, turkey, beef liver, bananas, avocados, tempeh
 - **Biotin:** Egg yolks, beef and chicken liver, dairy, mushrooms, legumes
 - **Folate:** Spinach, collard greens, mustard greens, asparagus, Brussels sprouts, lentils
- **Vitamin C:** Guava, strawberries, kiwis, tomatoes, red and green bell peppers
- **Magnesium:** Pumpkin seeds, chia seeds, flaxseeds, Brazil nuts, cashews, almonds, legumes, buckwheat, oats

Foundational Nutritional Supplementation

(if food-first approach doesn't replete deficiencies):

High-quality multivitamin with special attention to:

- **B-complex** (co-factors in hormone production)
 - B5 – pantothenic acid (100-150 mg)
 - B6 – pyridoxine (50-100 mg)
 - Biotin (1,000 mcg)
 - Folate (400-800 mcg)
- **Vitamin C** (1-2 grams) and antioxidant blend
- **Magnesium** (400-600 mg)

References

FOUNDATIONAL NUTRIENT SUPPLEMENTATION

- Eisenstein AB. Effects of dietary factors on production of adrenal steroid hormones. *Am J Clin Nutr.* 1957;5(4):369-376. doi:10.1093/ajcn/5.4.369

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- Moritz B, Schmitz AE, Rodrigues ALS, Dafre AL, Cunha MP. The role of vitamin C in stress-related disorders. *J Nutr Biochem.* 2020;85:108459. doi:10.1016/j.jnutbio.2020.108459

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- Cuciureanu MD, Vink R. Magnesium and stress. In: Vink R, Nechifor M, editors. Magnesium in the Central Nervous System [Internet]. Adelaide (AU): University of Adelaide Press; 2011. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK507250/>
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Foundational Therapeutic Approach: Chronic Stress & Adrenal Dysfunction/ Communication & Hormone Balance



LIFESTYLE FIRST

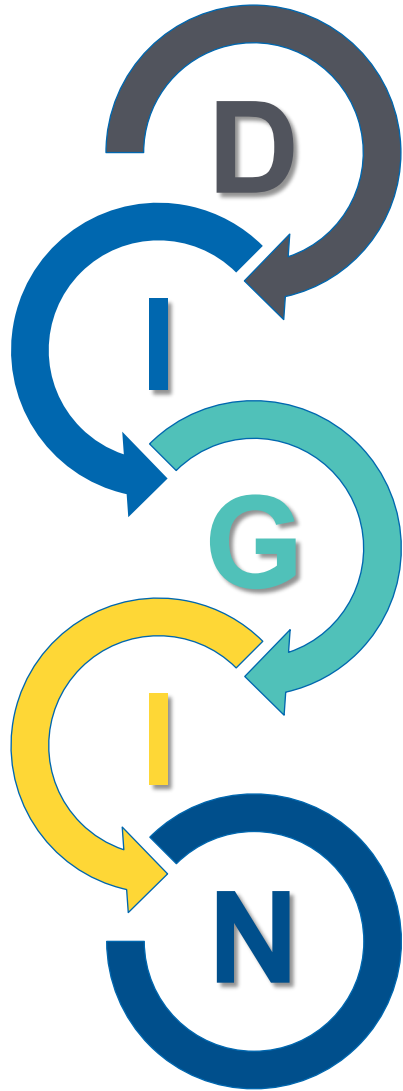


FOOD FIRST



GUT FIRST

The DIGIN Mnemonic of Gut Dysfunctions



Digestion/Absorption

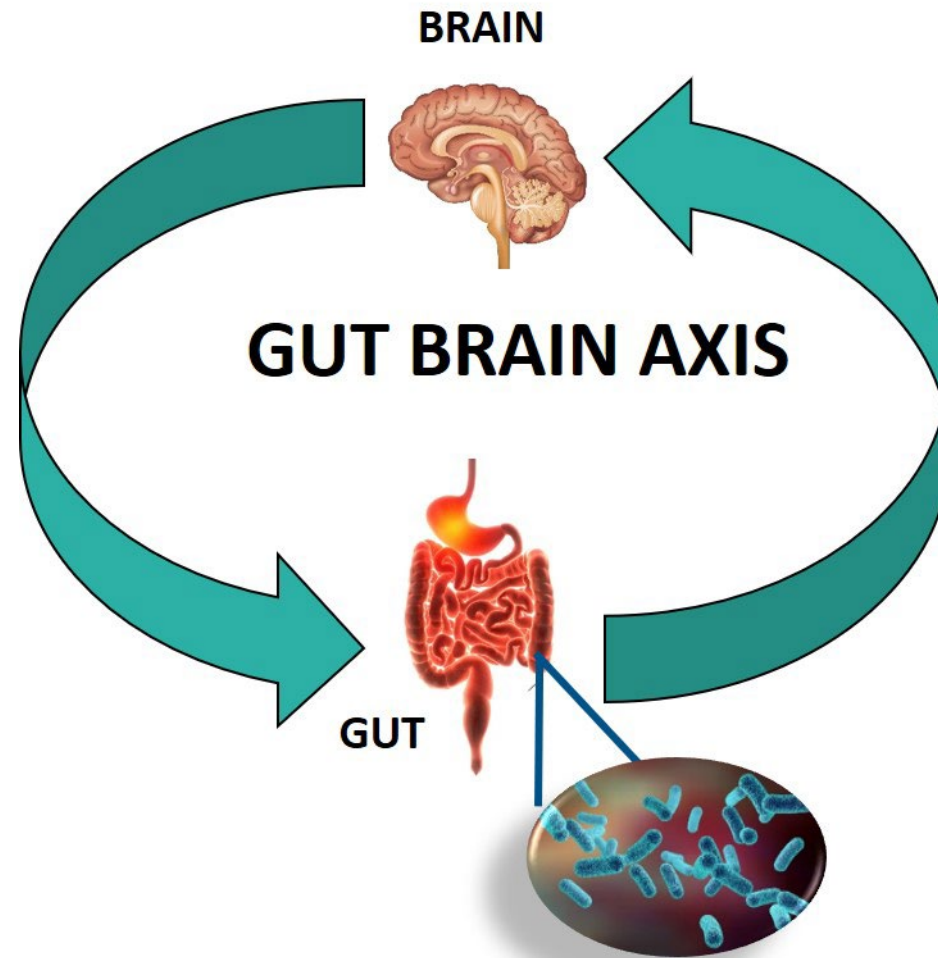
Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

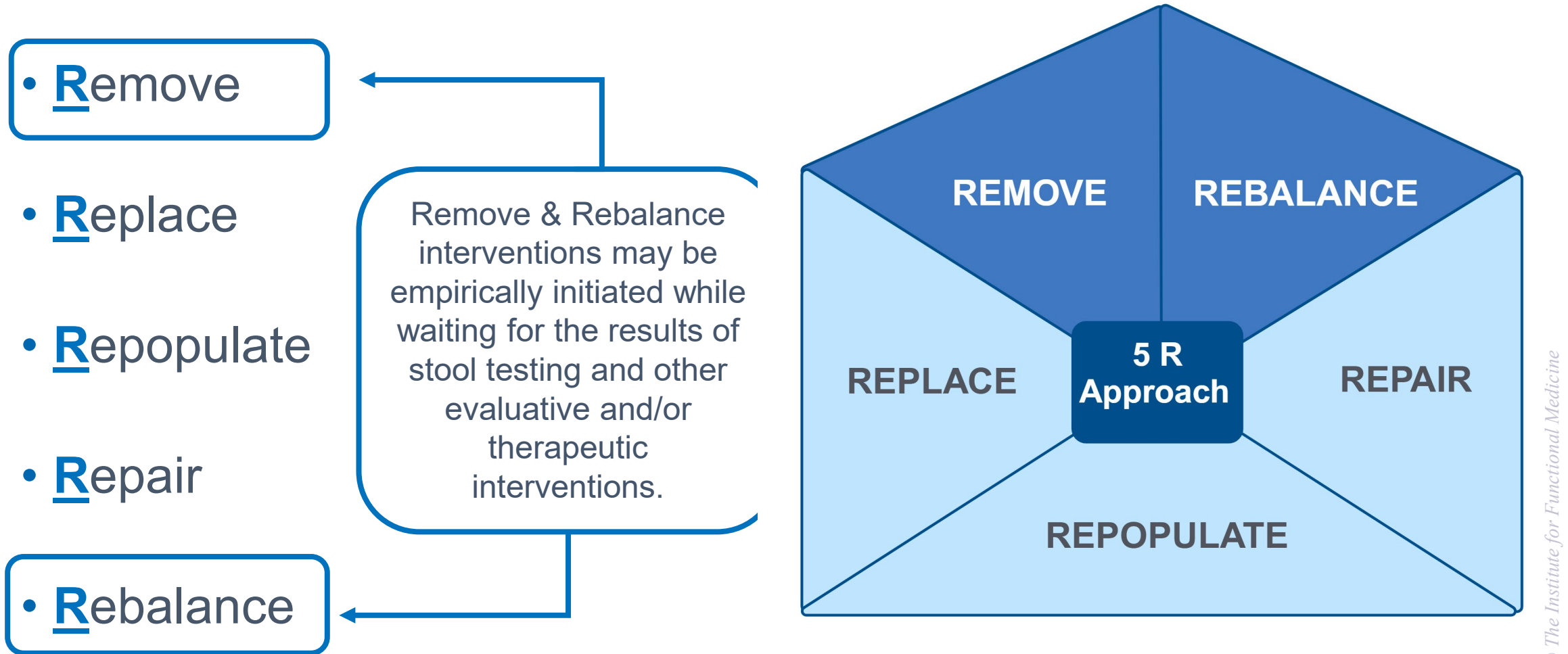
Nervous System

Gut-Brain Axis



Usual Sequence of the 5R Program

Primacy of the Remove and Rebalance Interventions



What gives the adrenal glands meaning and purpose?



Clinical Focus:

- Summary, conclusion, and key clinical takeaways.



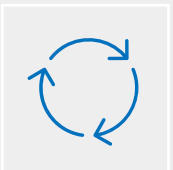
Adrenal Dysregulation: Clinical Takeaways



Most chronic diseases are associated with adrenal dysregulation, cortisol imbalance, and chronic stress.



Regardless of the nature and severity of adrenal dysregulation and cortisol imbalance, in most cases, it can be addressed with foundational dietary, lifestyle, and gut interventions without laboratory testing or supplementation.



Addressing each patient's personalized ATMs and matrix dysfunctions is essential for favorable therapeutic outcomes.

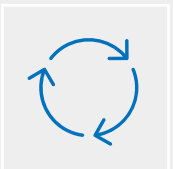
Adrenal Dysregulation: Clinical Takeaways



Rapid elevation in catecholamine and cortisol levels are part of the normal allostatic response to acute stress.

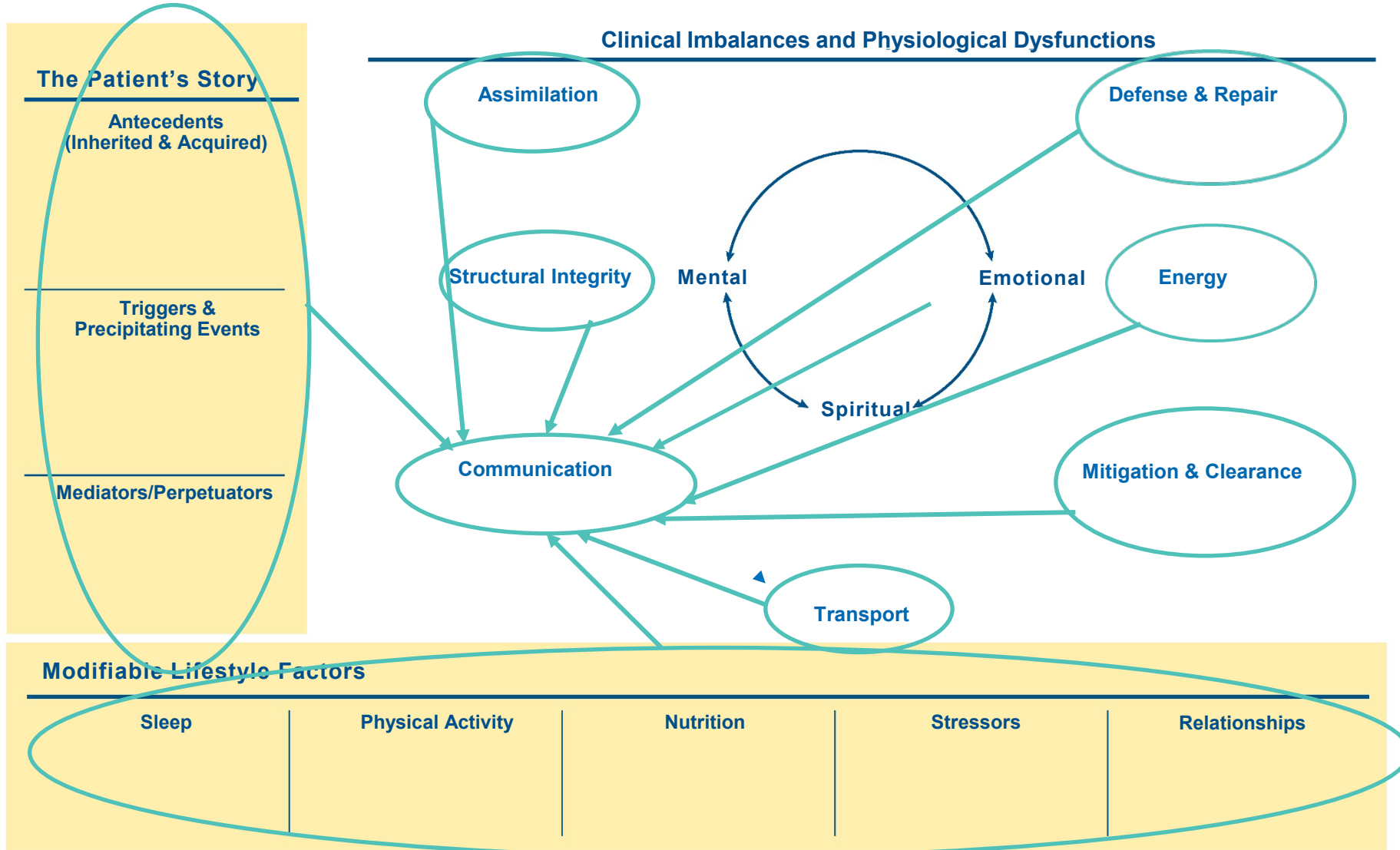


Chronic stimulation/upregulation of this allostatic response results in adrenal dysregulation and cortisol imbalance.



Chronic stressors are the most common mediators of adrenal dysregulation and must be therapeutically addressed.

Functional Medicine Matrix

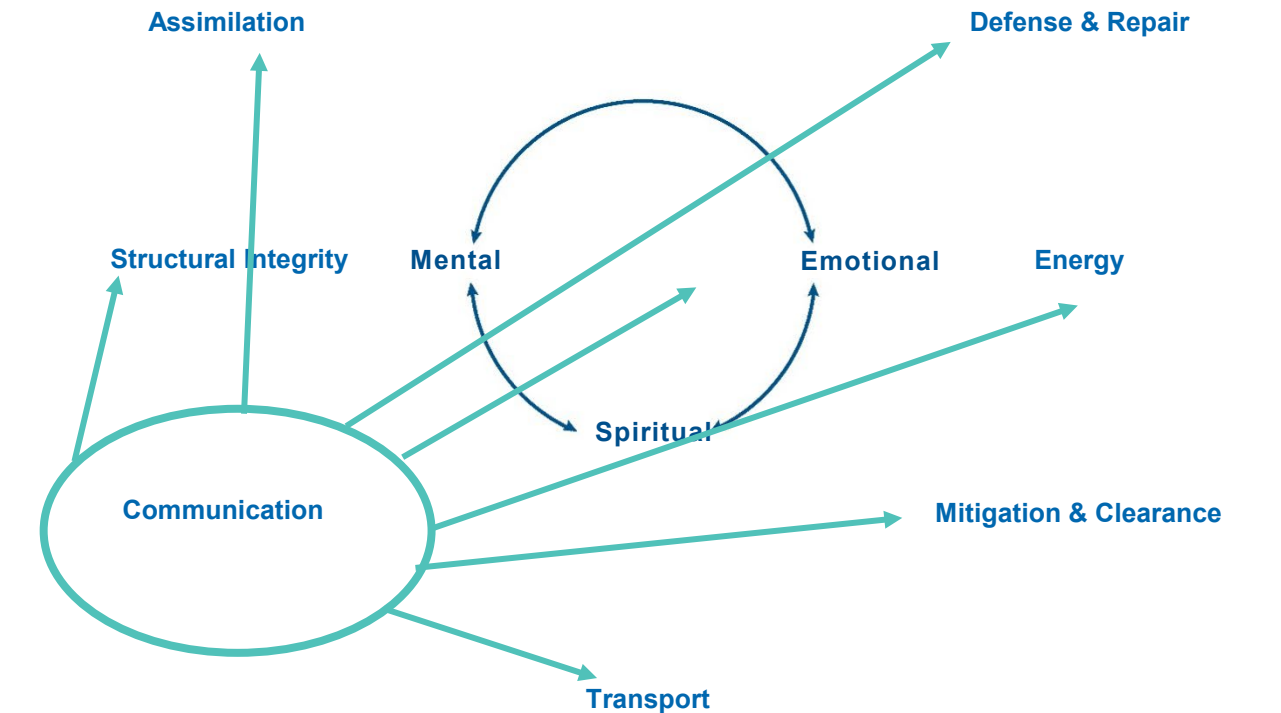


External and Internal Environmental Stressors that Contribute to Adrenal Dysregulation

Functional Medicine Matrix

Systemic Effects
of Adrenal
Dysregulation
and the Chronic
Stress Response

Clinical Imbalances and Physiological Dysfunctions



The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

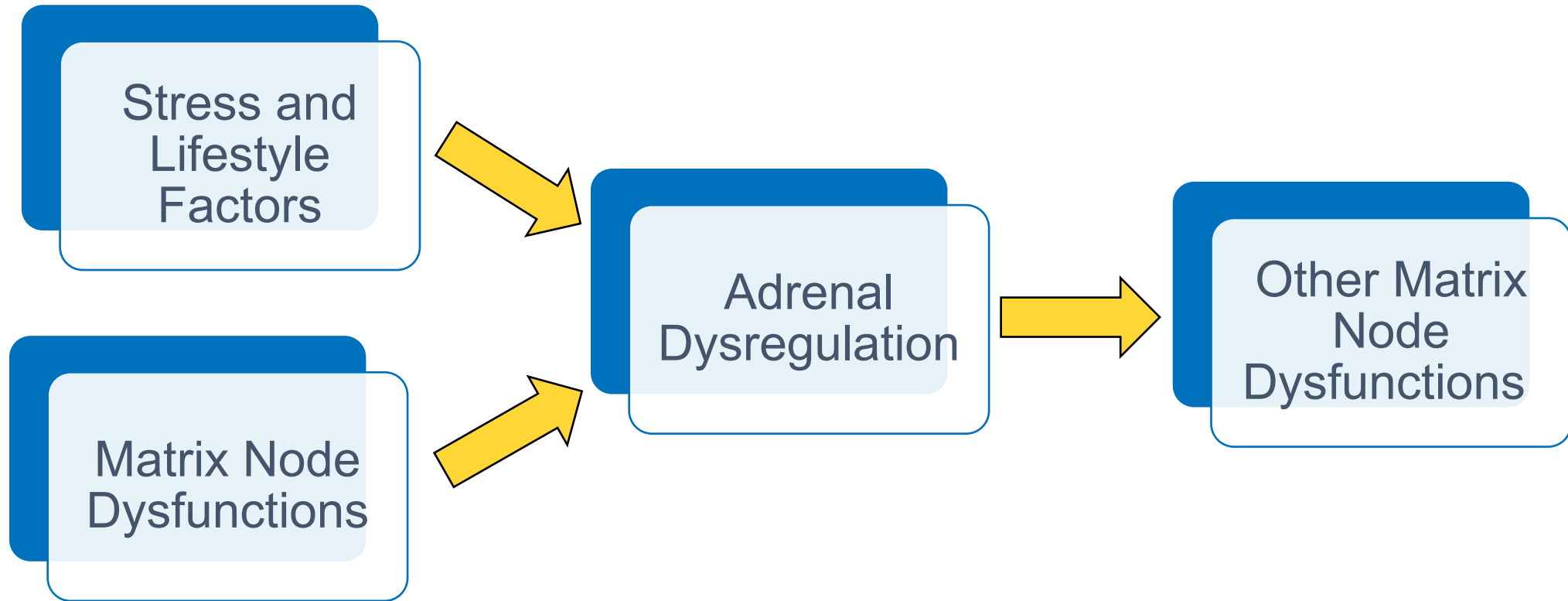
Nutrition

Stressors

Relationships

Complexity of Causative Relationships

Adrenal Dysregulation, Stress, Lifestyle, and Matrix Node Functions



- Allen JO, Mezuk B, Byrd DR, et al. Mechanisms of cardiometabolic health outcomes and disparities: what characteristics of chronic stressors are linked to HPA-axis dysregulation? *J Aging Health*. 2022;34(3):448-459. doi:10.1177/08982643221085903
- Le-Ha C, Herbison CE, Beilin LJ, et al. Hypothalamic-pituitary-adrenal axis activity under resting conditions and cardiovascular risk factors in adolescents. *Psychoneuroendocrinology*. 2016;66:118-124. doi:10.1016/j.psyneuen.2016.01.002

Long Description:

Complexity of Causative Relationships

Adrenal Dysregulation, Stress, Lifestyle, and Matrix Node Functions

- This flow diagram illustrates the interconnected causes of **adrenal dysregulation** and its downstream effects, particularly in relation to other matrix node dysfunctions.
- **Adrenal dysregulation** is influenced by:
 - **Stress and lifestyle factors**
 - **Matrix node dysfunctions**
- In turn, **adrenal dysregulation** contributes to other matrix node dysfunctions

Foundational Nutritional Supplementation

(if food-first approach doesn't replete deficiencies):

High-quality multivitamin with special attention to:

- **B-complex** (co-factors in hormone production)
 - B5 – pantothenic acid (100-150 mg)
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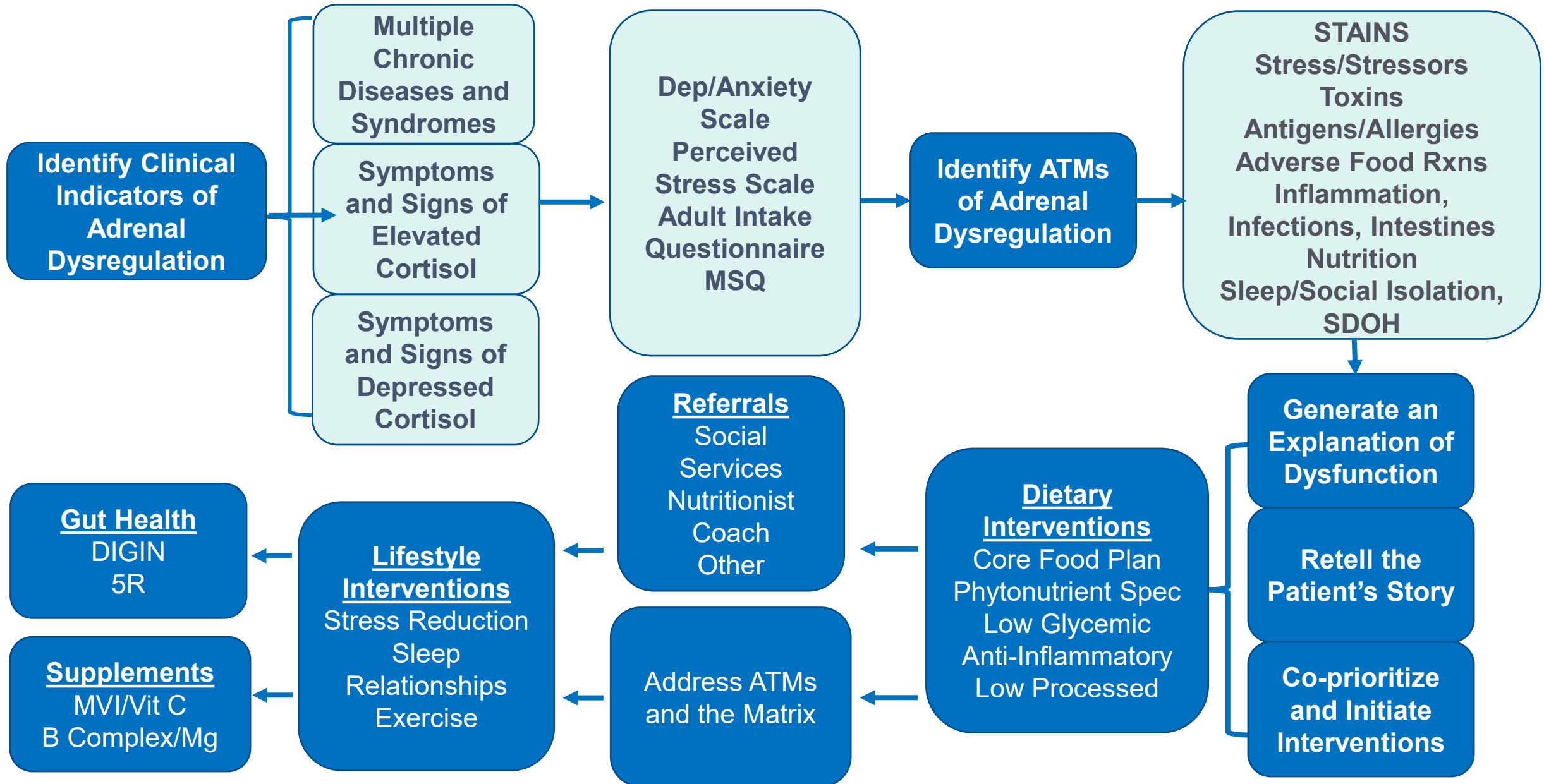
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Clinical Flow Path: Adrenal Dysregulation



Long description - Clinical Flow Path: Adrenal Dysregulation

This clinical flow diagram outlines a step-by-step approach to identifying and managing **adrenal dysregulation** through assessment and targeted interventions.

Step 1: Identify Clinical Indicators of Adrenal Dysregulation

- Consider the presence of:
 - Multiple chronic diseases and syndromes
 - Symptoms and signs of elevated cortisol
 - Symptoms and signs of depressed cortisol

Step 2: Use Assessment Tools

- Use the following questionnaires:
 - Depression/Anxiety Scale
 - Perceived Stress Scale
 - Adult Intake Questionnaire
 - Medical Symptoms Questionnaire (MSQ)

Step 3: Identify A T Ms (Antecedents, Triggers, Mediators)

- Use the **S T A I N S** acronym to guide identification:
 - Stress/Stressors
 - Toxins
 - Antigens/Allergies
 - Adverse Food Reactions
 - Inflammation
 - Infections
 - Intestinal dysfunction
 - Nutrition
 - Sleep/Social Isolation
 - Social Determinants of Health

Step 4: Generate an Explanation of Dysfunction

- Develop a clinical explanation
- Retell the patient's story
- Co-prioritize and initiate interventions

Step 5: Implement Personalized Interventions

- **Address A T Ms and the Matrix** through the following:
 - **Referrals** to social services, nutritionists, health coaches, or others
 - **Dietary Interventions:** Core food plan, phytonutrient spectrum, low glycemic, anti-inflammatory, low processed foods
 - **Lifestyle Interventions:** Stress reduction, sleep, healthy relationships, exercise
 - **Supplements:** Multivitamin, vitamin C, B complex, magnesium
 - **Gut Health Support:** DIGIN protocol, 5R framework

Communication and Hormone Balance

Addressing Adrenal Dysregulation and Chronic Stress

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE